

Jeric Cantos | Cat Gipetto

# TravelTide Customer Segmentation

Version 1

# Agenda

- Introduction
- Methodology
- Personas / Segments
- Next Steps

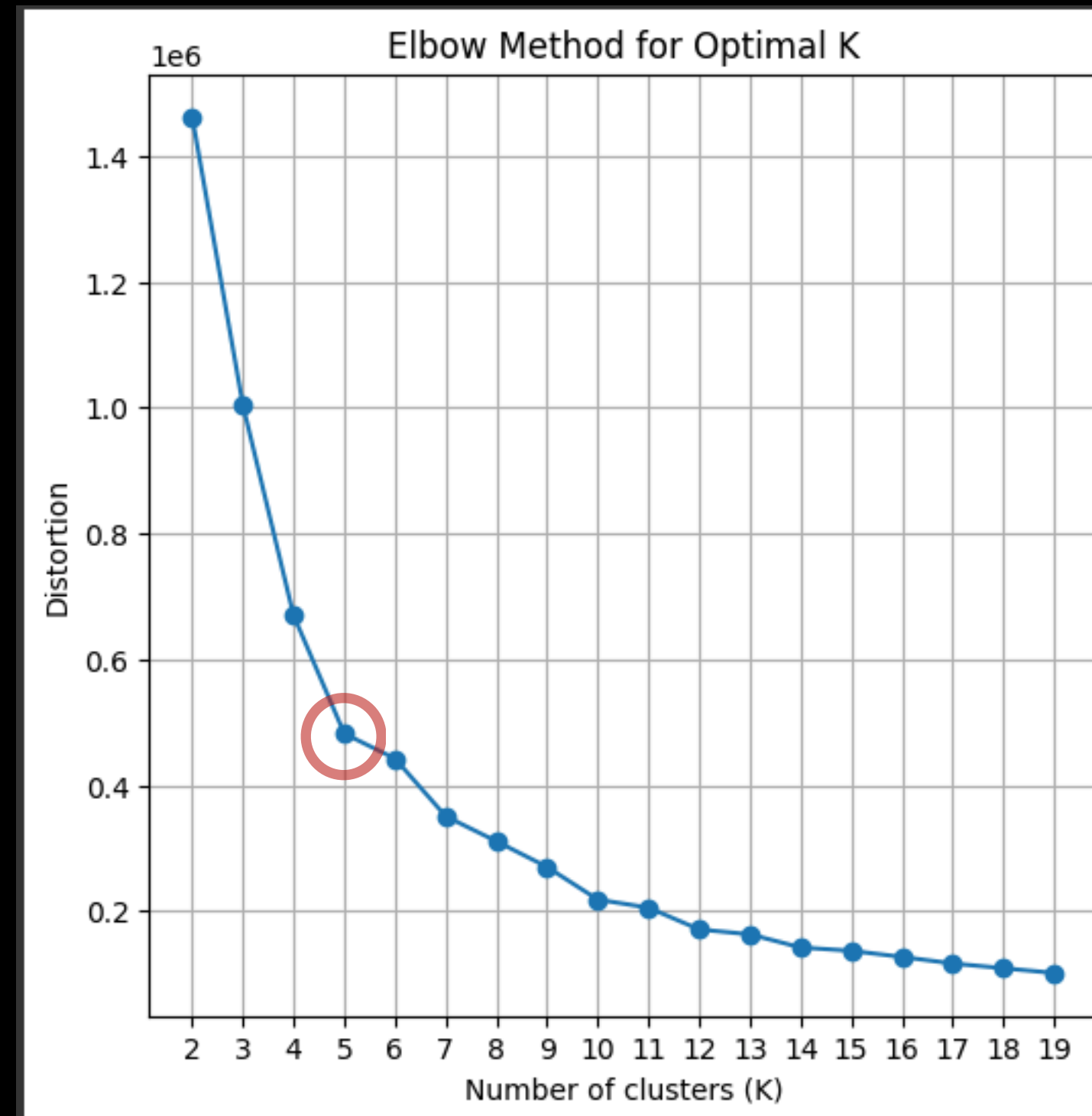


# Methodology

# ML Clustering

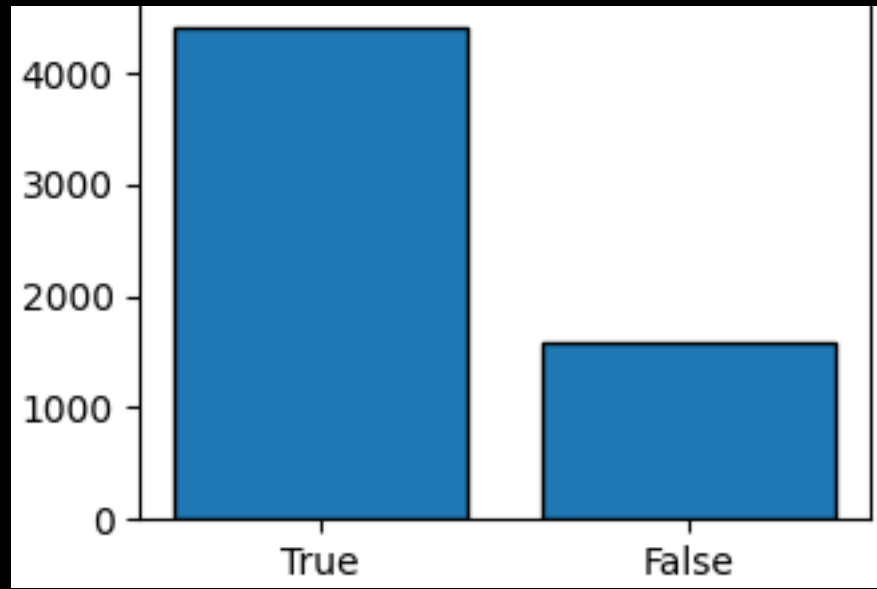
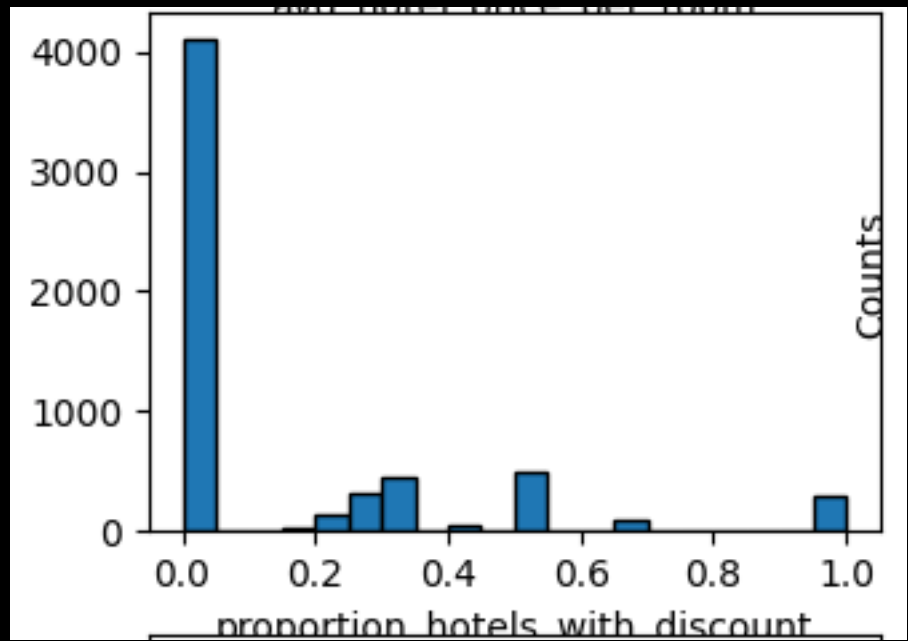
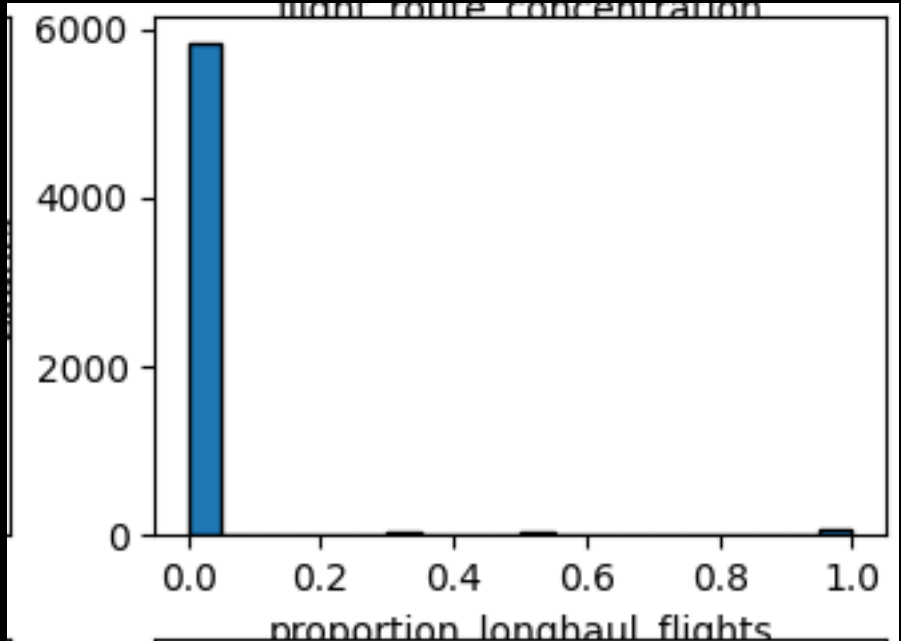
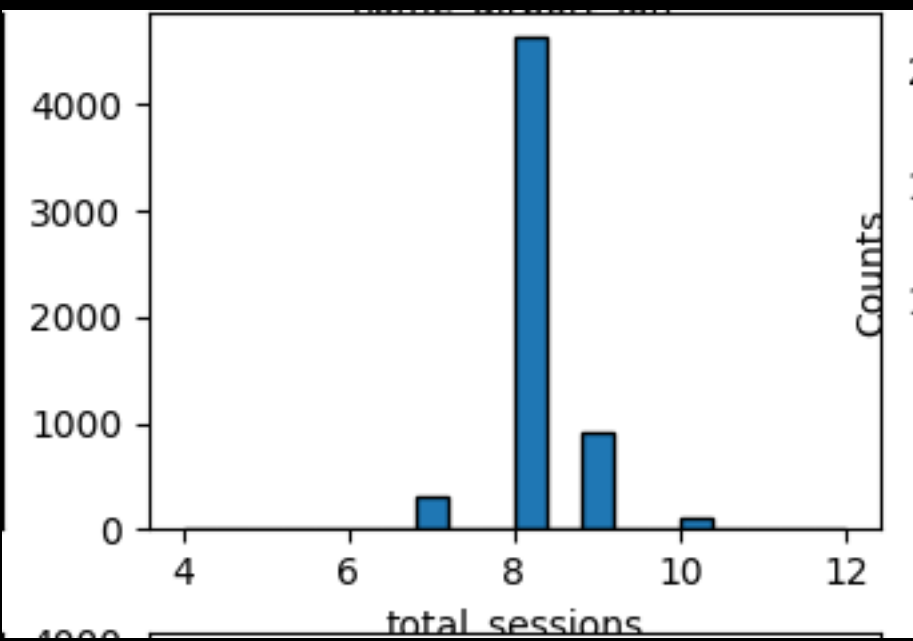
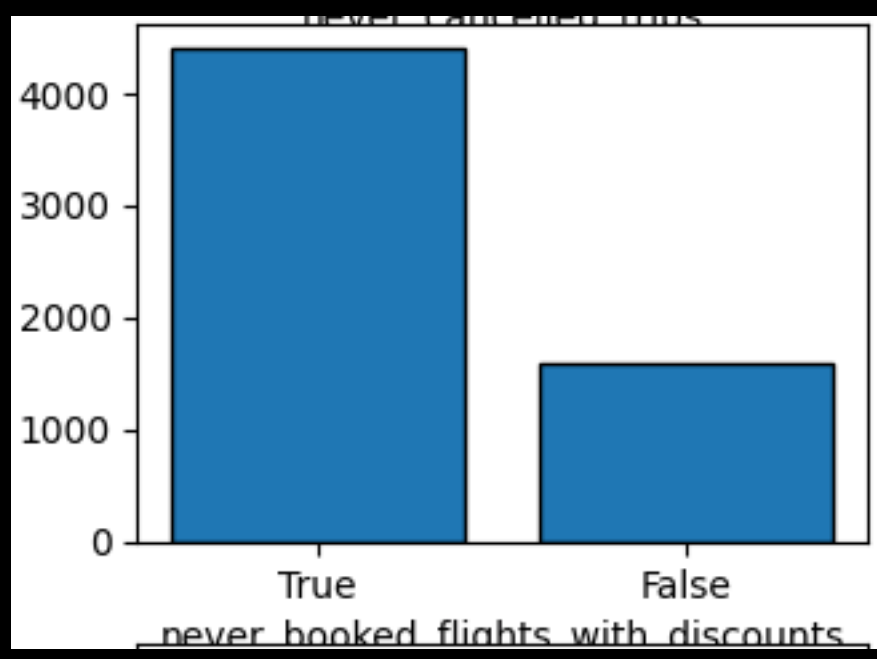
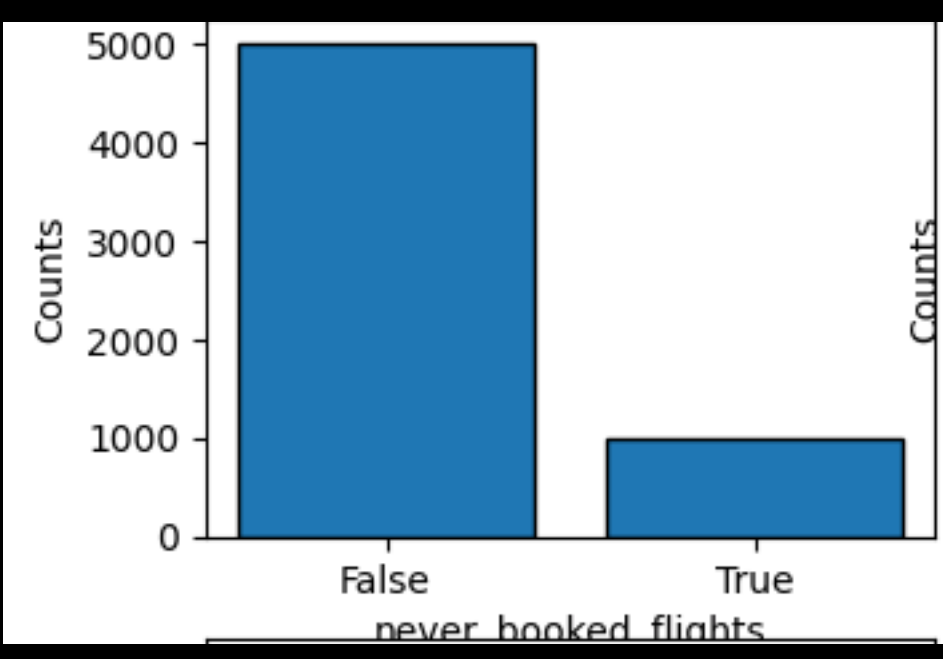
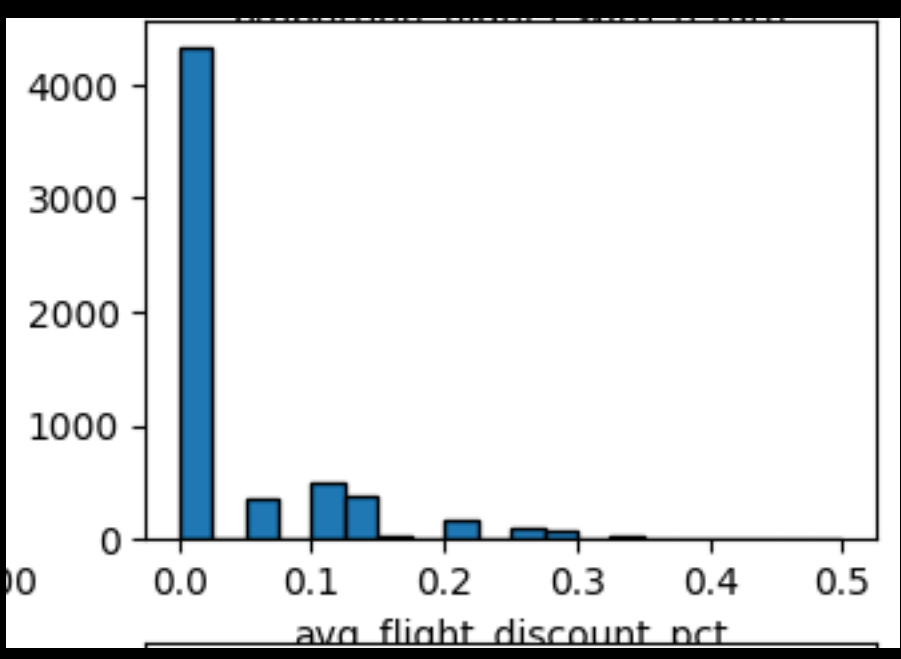
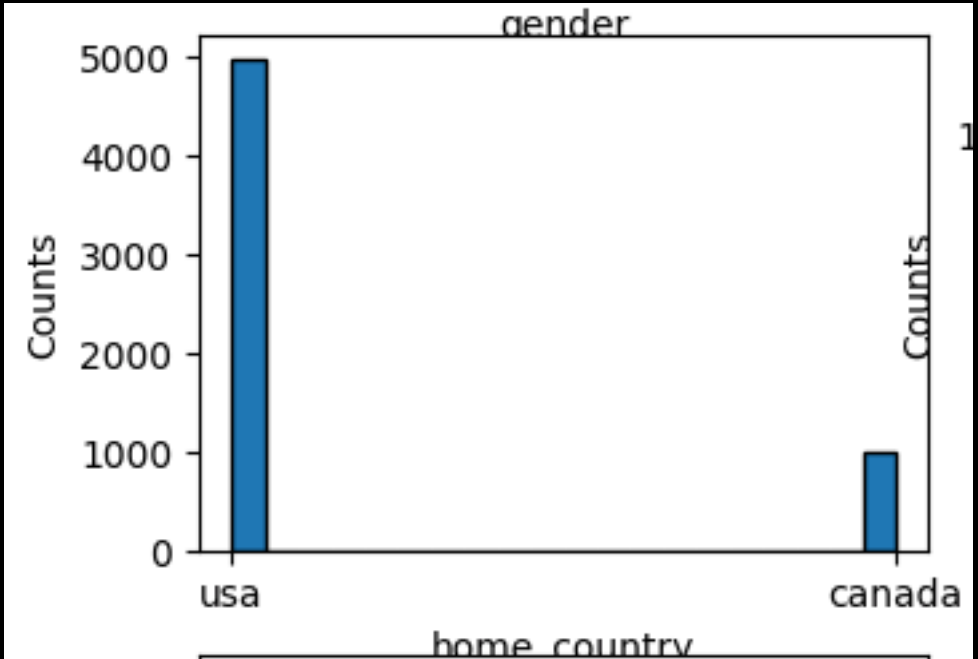
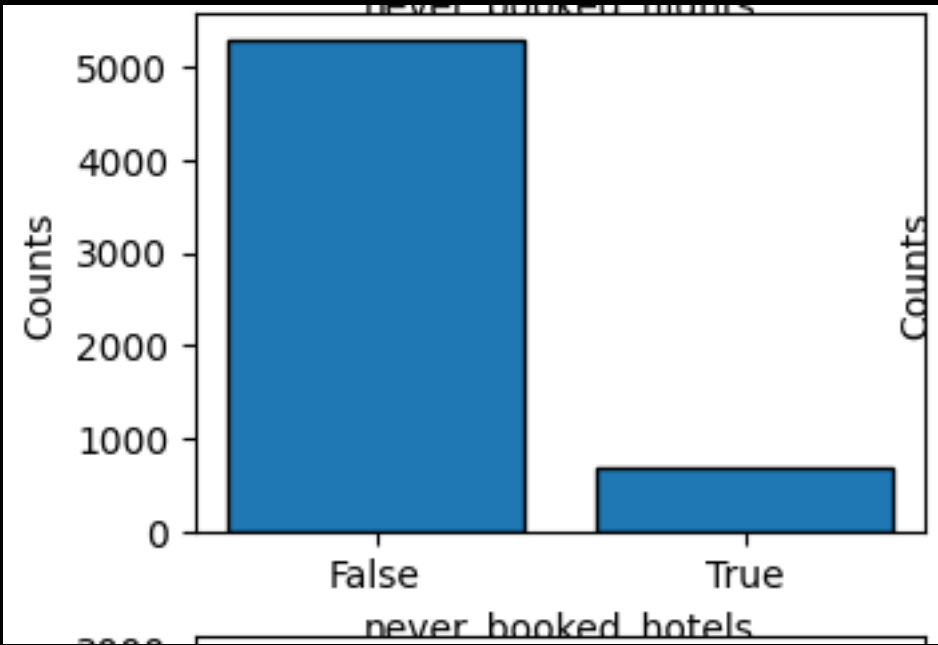
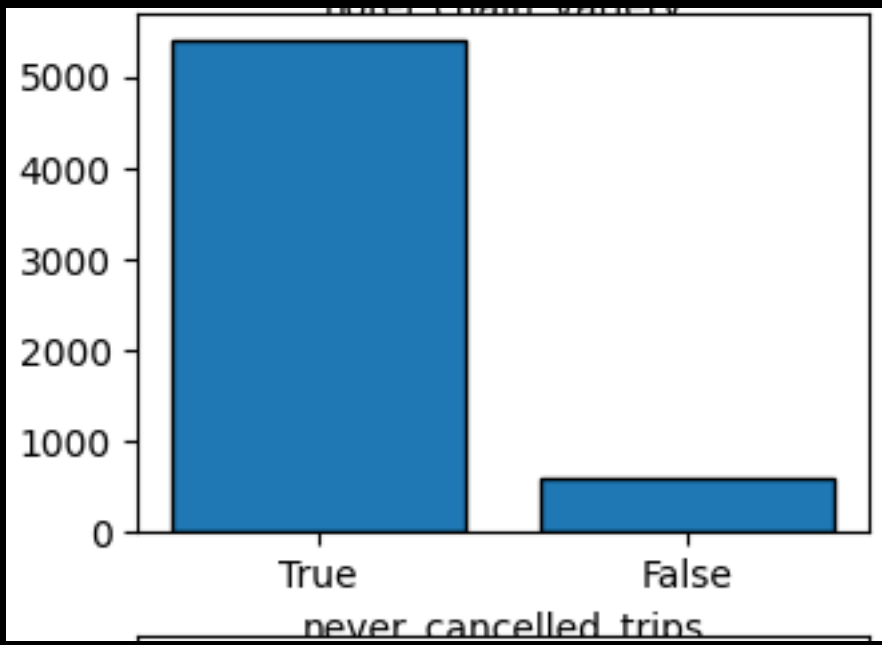
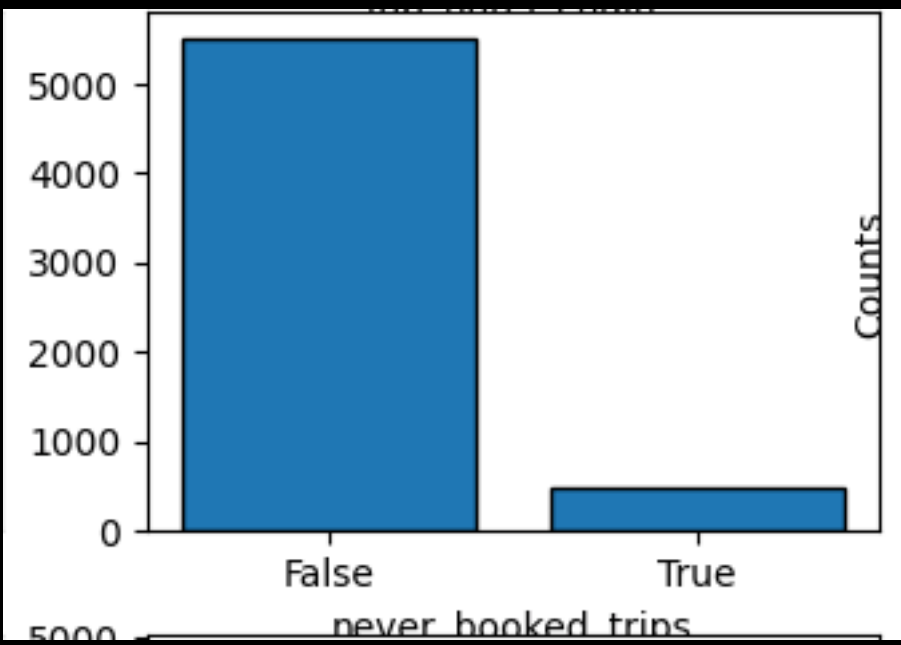
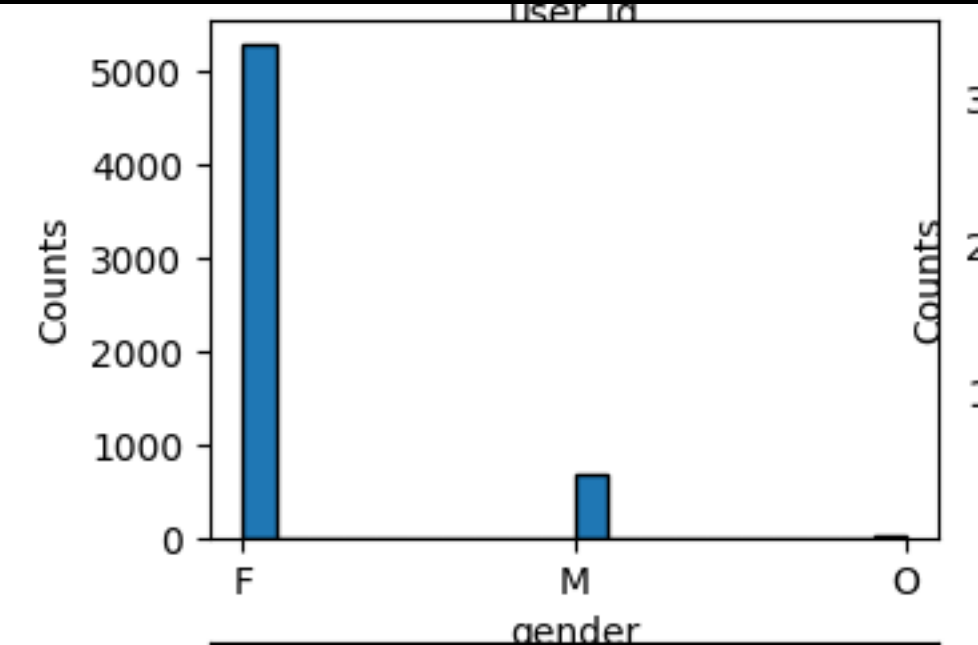
## KMeans

| col_0 | count |
|-------|-------|
| group |       |
| 0     | 721   |
| 1     | 123   |
| 2     | 144   |
| 3     | 119   |
| 4     | 4723  |



- 5 groups - Elbow with highest silhouette score (0.72)
- Dominated by one class

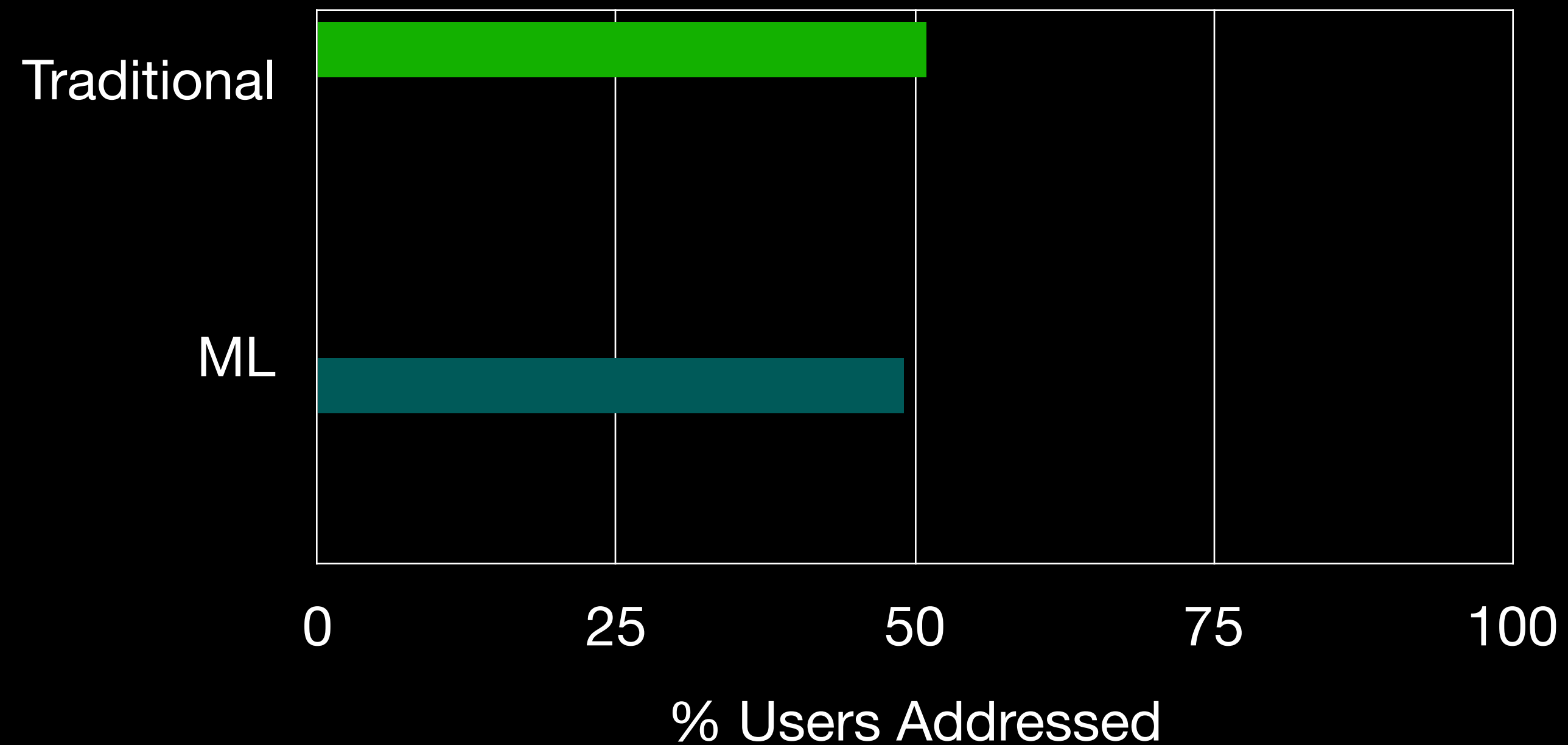
# Homogenous Dataset



# Hybrid Approach

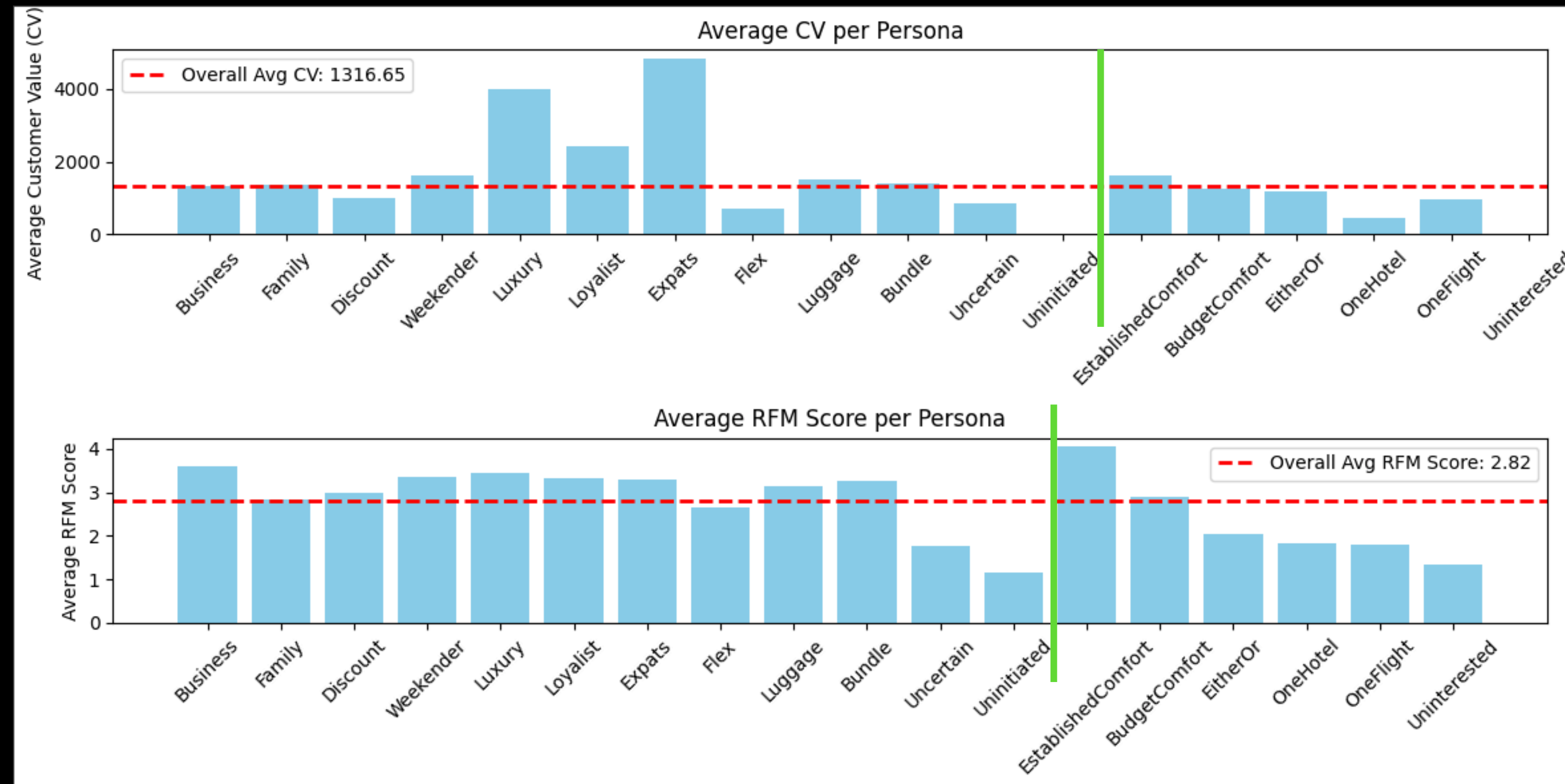
# Traditional and ML Segmentation

Machine discovers the unaddressed



# Traditional and ML Segmentation

## Machine discovers the unaddressed



# Persona Categorization

Strong - Growth - Weak

|             | Strong<br>Perks based on motivations                                                                                                                                                          | Growth<br>Perks based on upsell potential        | Weak<br>Lower priority               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------|
| Traditional | Business Traveler<br>Family Traveler<br>Discount Hunter<br>Weekender Couple<br>Luxury Traveler<br>Premium Loyalist<br>Expatriate<br>Flexibility-first Planner<br>Luggage-heavy Frequent Flyer | Bundle Saver<br>Uncertain Shopper<br>Uninitiated |                                      |
| ML          | Established Comfort Seeker<br>Budget Comfort Seeker                                                                                                                                           | Bundle Saver Lite<br>One-off Flyer               | One-off Hotel Booker<br>Uninterested |

# Strong Personas

Traditional Segmentation



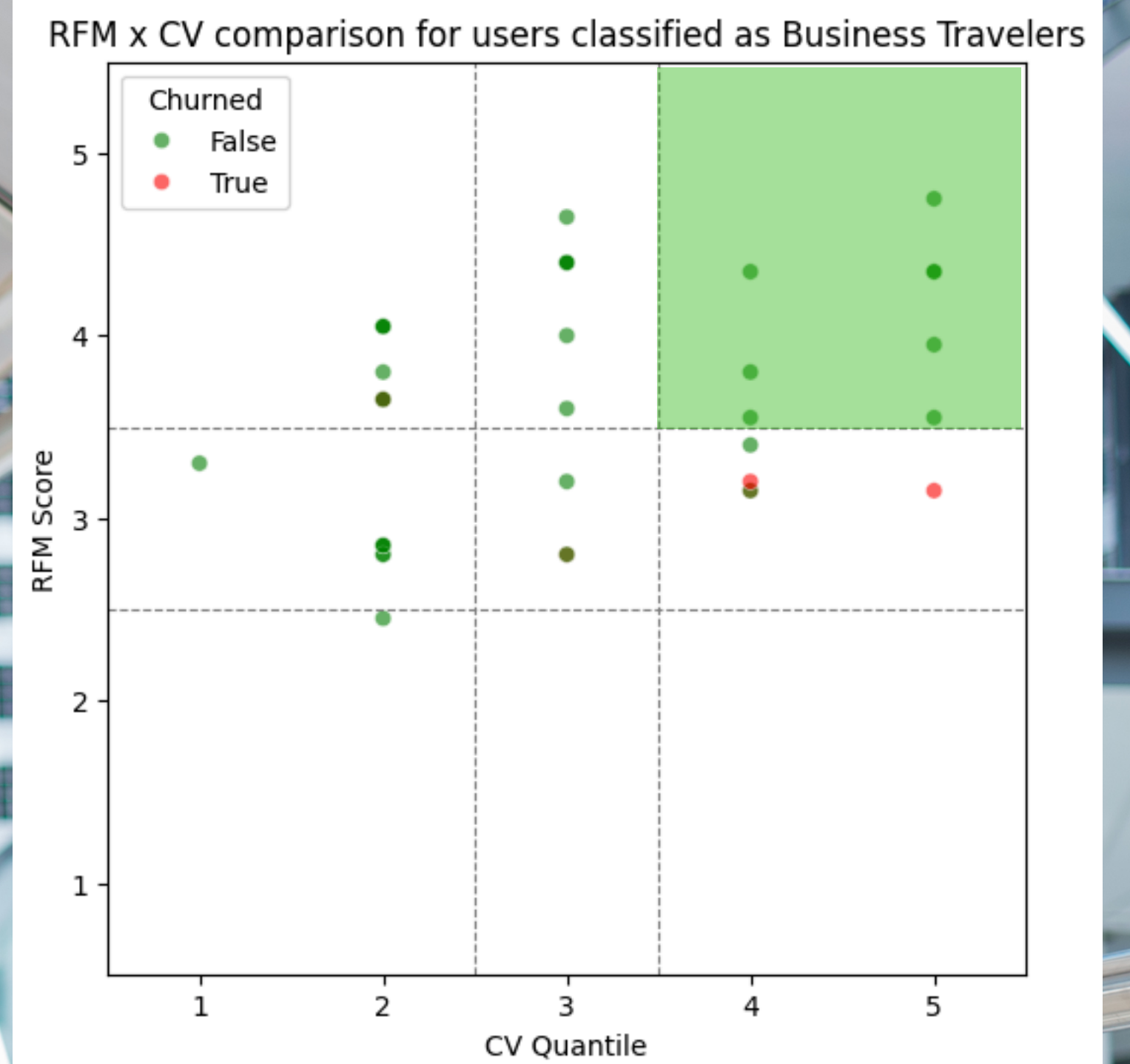
Numerous trips  
Same destinations  
Same airlines  
Above average spending on flights  
and hotels

# Business Travelers

Flexibility and Comfort

# Business Travelers

36 Users



## VIP:

- Goal: retain
- Lounge access, loyalty bonuses, free cancellation

## Core:

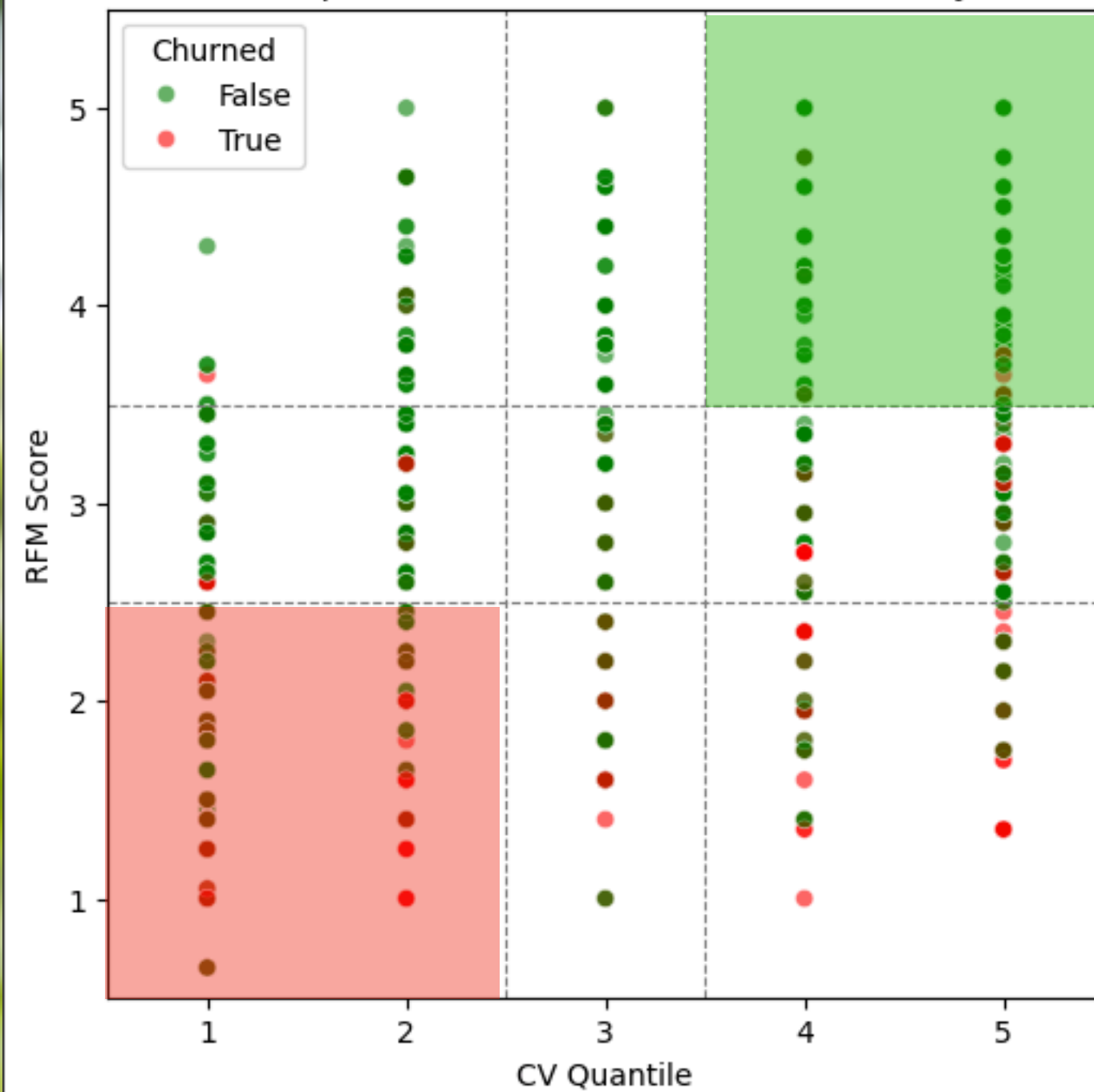
- Goal: improve spending
- Exclusive discounts to make more expensive options more attractive, free cancellation

Longer than average trip lengths  
Has children or books two rooms or  
more (because they have children)

# Family Travelers

Cost-effective fun with kids

RFM x CV comparison for users classified as Family Travelers



## VIP:

- Goal: retain / recover churned
- 1 Free hotel meal
- Family-friendly features e.g. early/late check-in/out, priority boarding, discount on airport-hotel transfers
- 1 night free hotel with flight (minimum X nights) for churned.

## Core:

- Goal: nudge to higher levels
- Free hotel meal
- 1 night free hotel with flight (minimum X nights, Y seats, Z rooms)
- Upsell bundles (flight+hotel bundles at deeper discount, or more expensive bundles seats/hotels with bundled hotel-airport transfers or family activities)

## At Risk:

- Goal: Low-cost activation
- Credit Wallet (e.g. 50 EUR off next booking)
- Free cancellation, price match guarantee.

# Family Travelers

1040 Users

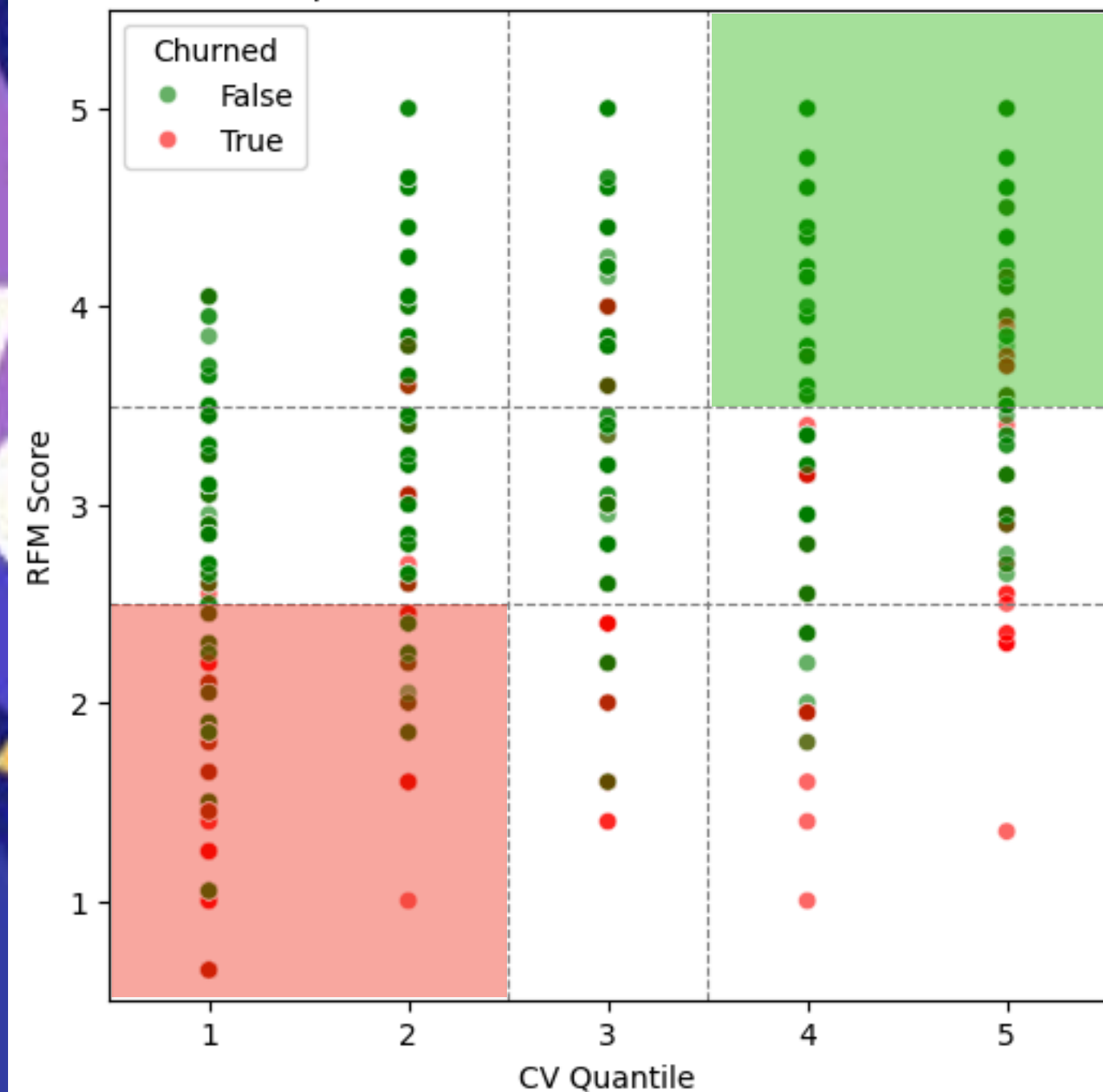
Lower average fare / room rate  
Higher average clicks per session  
because they are looking for deals

## “Best Deal” perception

# Discount Hunters

1114 Users

RFM x CV comparison for users classified as Discount Hunters



## VIP:

- Goal: sustain loyalty with minimal cost
- Premium deal bundles at discounted rates

## Core:

- Goal: improve activity i.e. volume over luxury
- Flash sales, price match guarantee

## At Risk:

- Goal: Low-cost activation
- Personalized deal alerts / reminder emails

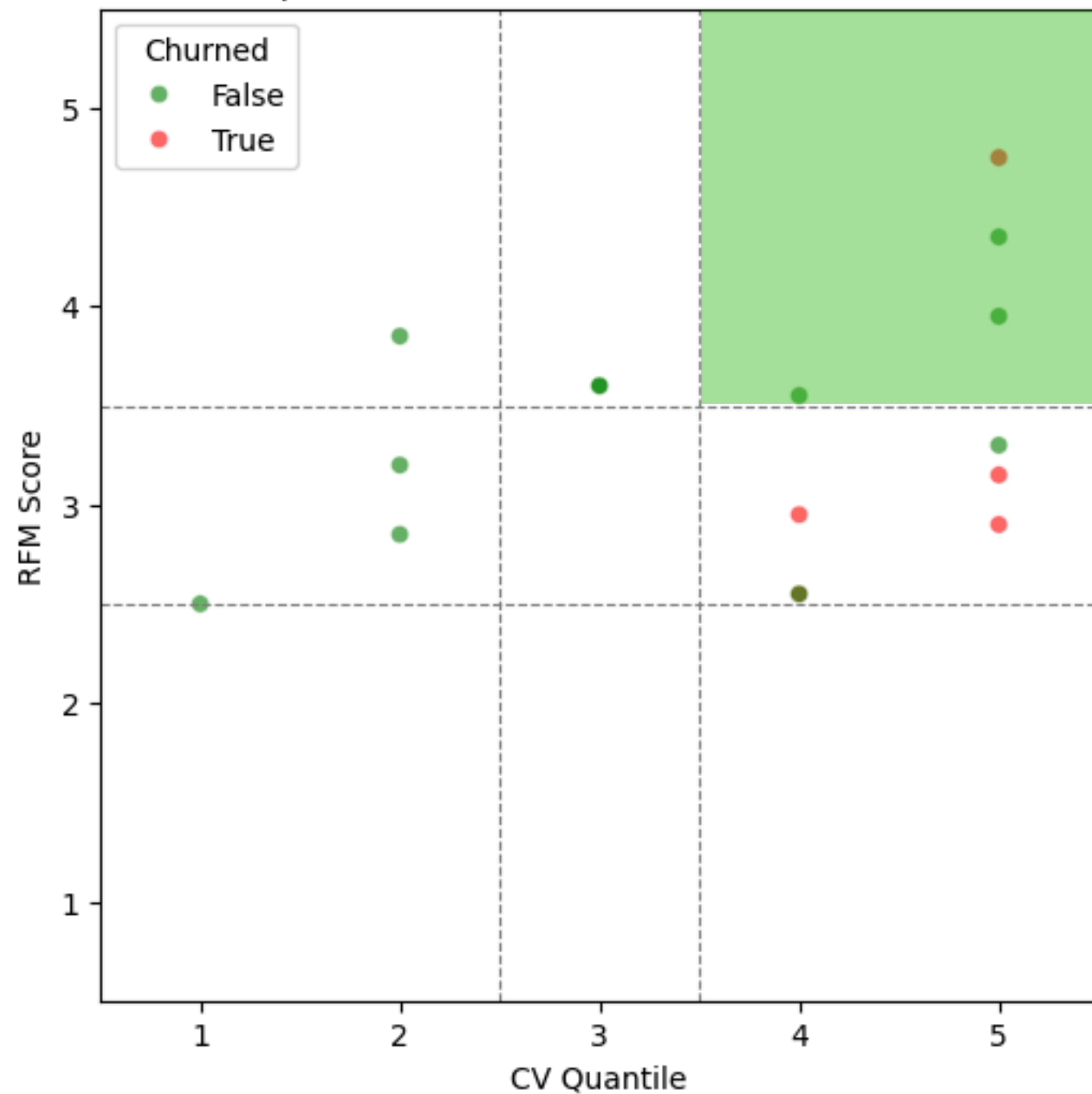
A romantic couple is seen from behind, sitting in a hammock strung between two large palm trees. The scene is set at sunset, with a warm, golden-orange glow in the sky. The couple is silhouetted against the bright light. The man is wearing a plaid shirt, and the woman is wearing a patterned top. The hammock is made of a dark fabric with a light-colored pattern. The palm trees are large and have fronds that hang down. The overall mood is peaceful and intimate.

Books 1 room, 2 seats.  
Shorter trip duration  
High variety of destinations  
Shorter booking lead time

# Weekender Couples

Quick experiences

RFM x CV comparison for users classified as Weekender Couples



### VIP:

- Goal: reward and retain
- VIP Weekend Packages (e.g. room upgrade, late check-out)
- Free hotel breakfast

### Core:

- Goal: improve spending / frequency
- “Next Trip” discount voucher
- Free couple activity (e.g. spa, bike rental) on 2nd trip in a period (e.g. month)

# Weekender Couples

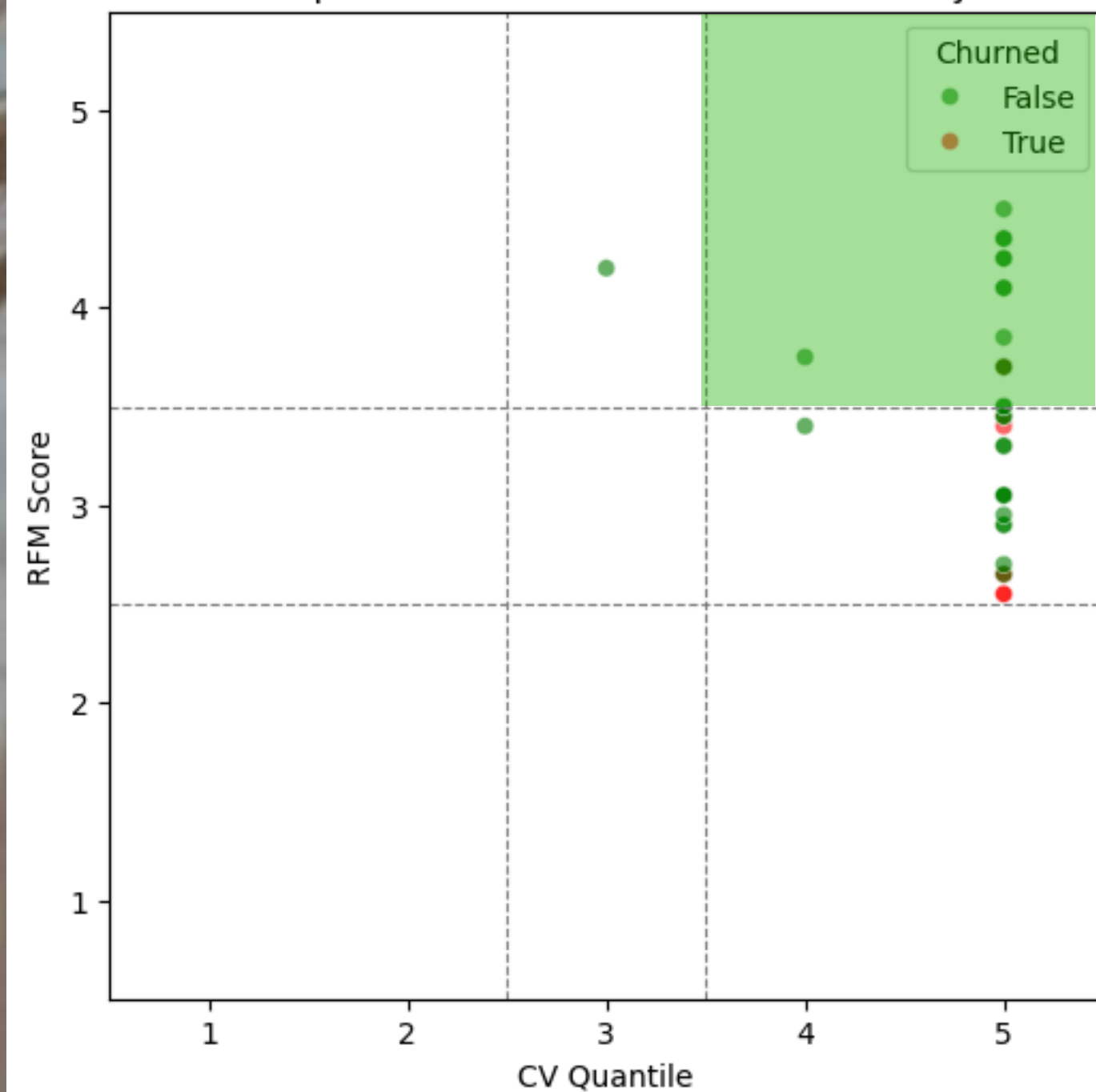
16 Users

Booked more than one trip  
High average flight and hotel price  
High total spend  
Low discount use  
Low average clicks because they  
are not too concerned about finding  
the best deals.

# Luxury Travelers

Exclusivity with no Friction

RFM x CV comparison for users classified as Luxury Travellers



## VIP:

- Goal: reward and retain
- Invite-only elite status
- Dedicated concierge, chauffeur for transfers, private gifts at hotel.

## Core:

- Goal: cement loyalty, improve consistency
- Dedicated concierge, guaranteed suite upgrade, exclusive event invites

# Luxury Travelers

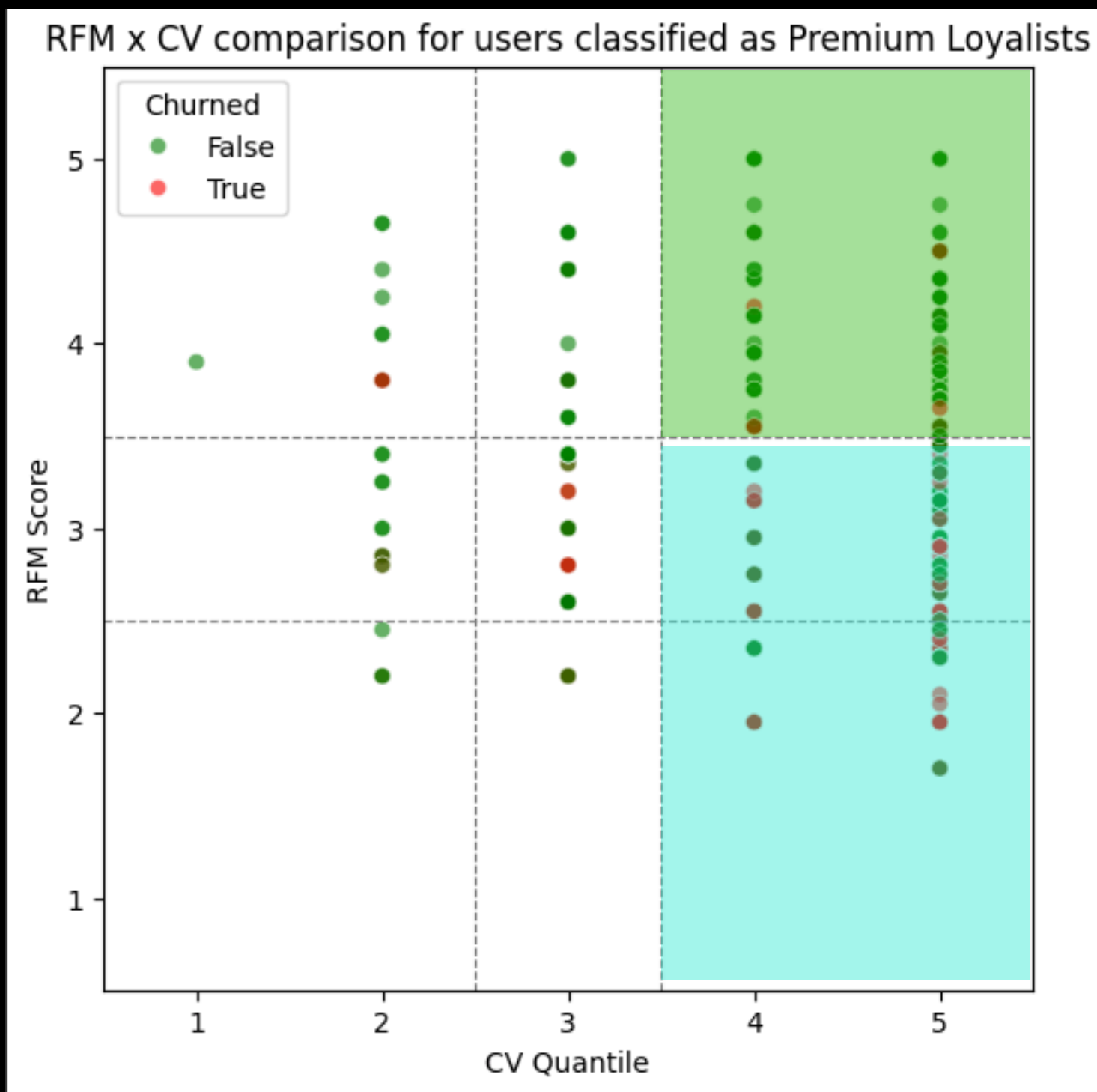
35 Users

Higher total spend  
Higher average flight / hotel price  
Loyal to airline and/or hotel chain  
Low trip cancellation rate



# Premium Loyalists

Reward brand loyalty



### VIP:

- Goal: cement top loyalty
- Tie-up with loyalty program providers to upgrade status (e.g. gold to platinum / platinum to invite-only tier)

### At Risk:

- Goal: Reactivate churned users, promote more consistency
- Temporary reinstatement of premium status for certain time period (e.g. reactivated Gold status for next 3 months, book twice to keep until end of year)

# Premium Loyalists

434 Users

A central globe is surrounded by a collage of global landmarks including the Eiffel Tower, Big Ben, the Leaning Tower of Pisa, and the Sydney Opera House. Several commercial airplanes are shown in flight around the globe, and a yellow and blue high-speed train is visible at the bottom right. The background is a light beige gradient.

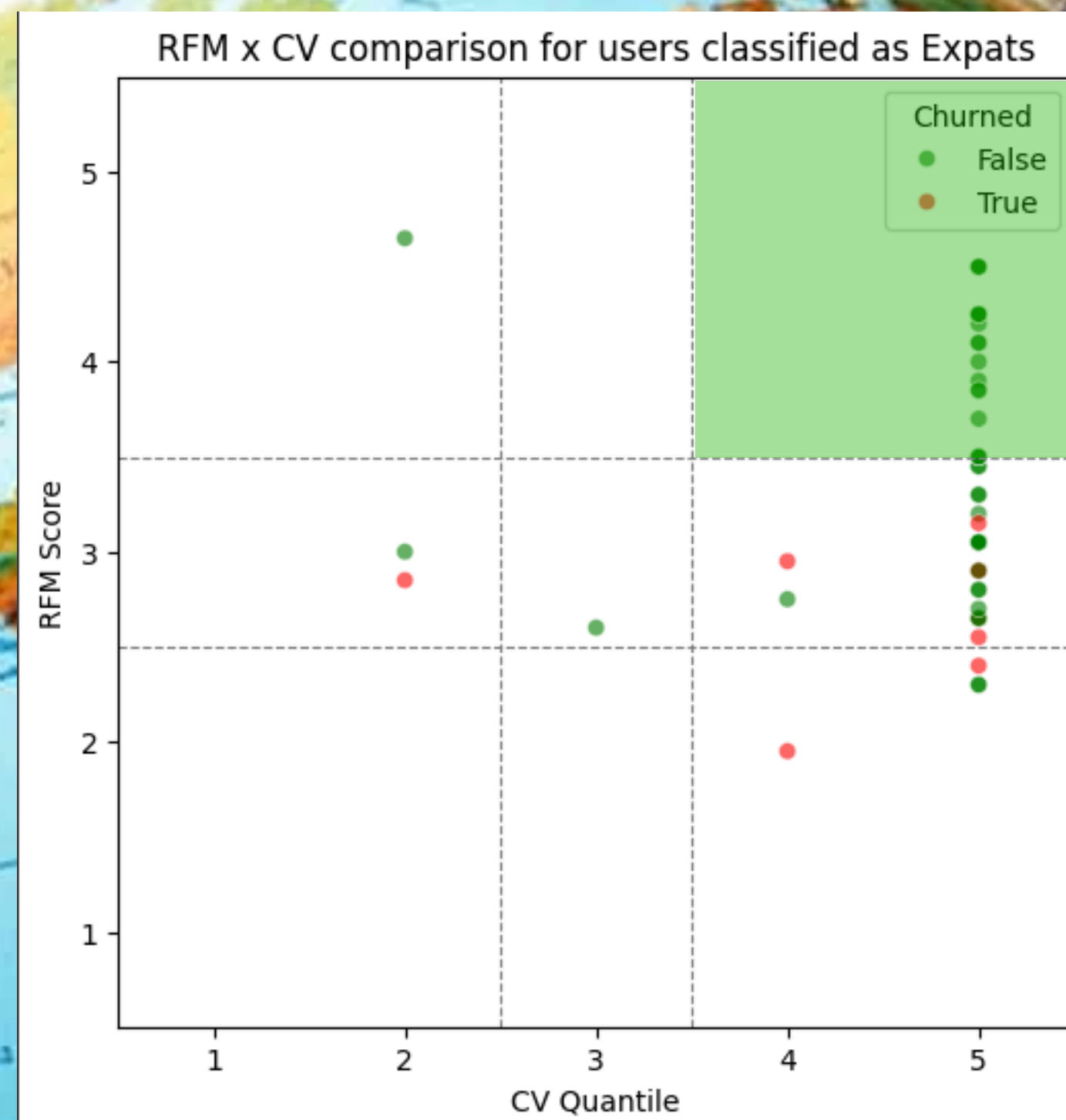
High proportion of long-haul flights  
Low flight route variance  
Longer trip duration e.g. two weeks  
Regularly travels with checked bags  
Longer booking lead time

# Expats

Cheap and comfy ride home

### VIP:

- Goal: cement top loyalty
- Get more value from their “miles flown” e.g. Business Class meal or legroom upgrade for long-haul flight.
- Early bird booking discounts
- Free checked bag



### Core:

- Goal: recover from churn, improve consistency
- Early bird booking discounts
- Free checked bag

# Expats

59 Users

A man with short dark hair, wearing a blue tank top and black shorts, is performing a yoga pose on a black mat on a green lawn. He is in a side plank variation, with his left leg bent and his right leg extended. His hands are clasped together near his right knee. The background is a lush green lawn with some white flowers in the distance.

Higher average booking lead time  
Non-zero cancelled trips  
Fairly active on the platform

# Flexibility-First Planners

Security in Uncertainty

# Flexibility-First Planners

432 Users

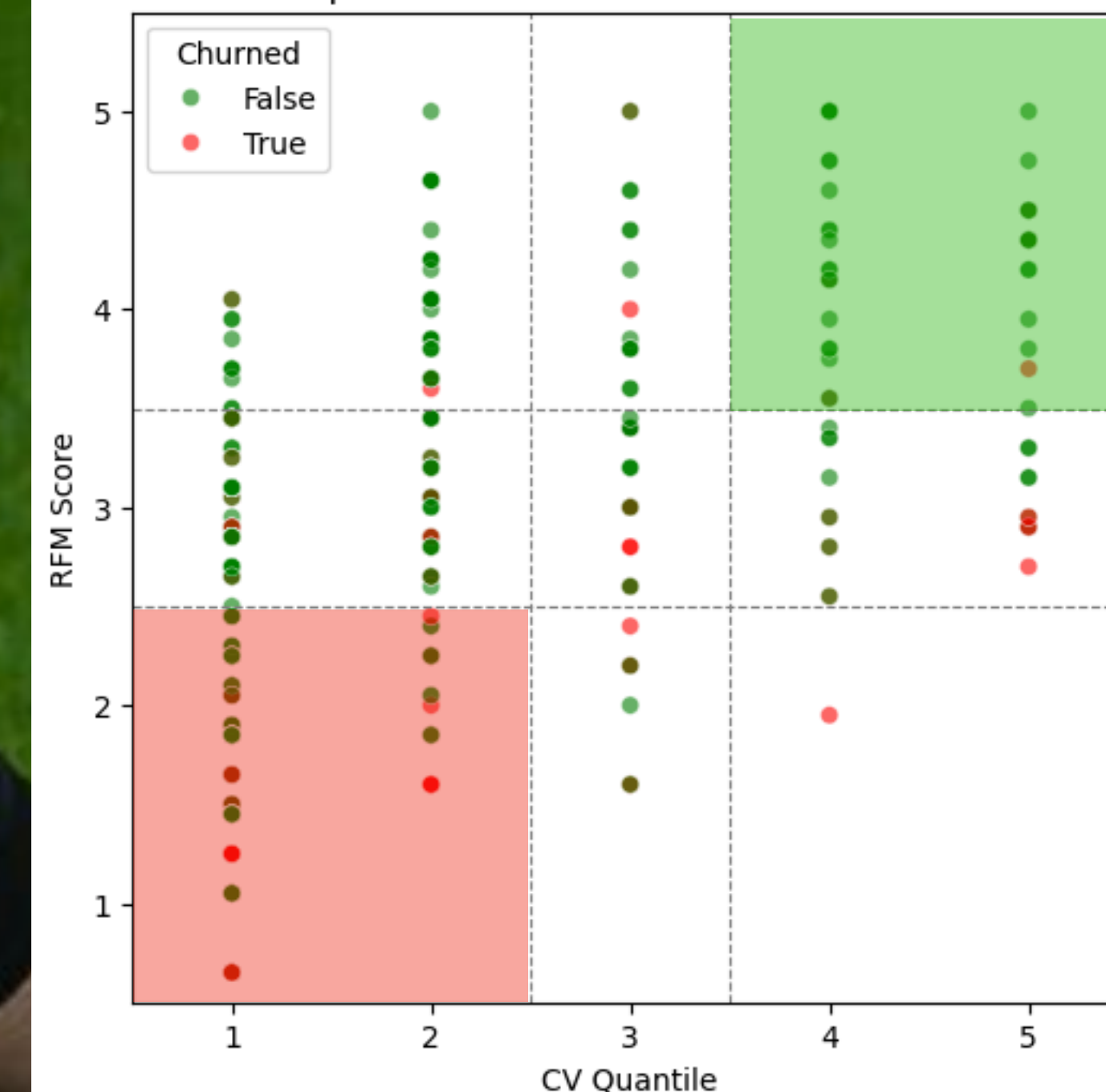
## Core:

- Goal: try to move to higher tier
- Free cancellation within 24-48hrs, free travel insurances if booking higher cost flights or hotels, or as perk for Nth booking in a period.

## VIP:

- Goal: provide maximum sense of security
- Free cancellation, inclusion of travel insurances

RFM x CV comparison for users classified as Flex-first Planners



## At Risk:

- Goal: Low-cost activation / deprioritize
- Deal alerts / reminder emails



Had booked at least two flights  
Checked at least one bag on  
average

**Luggage-heavy Frequent Flyers**

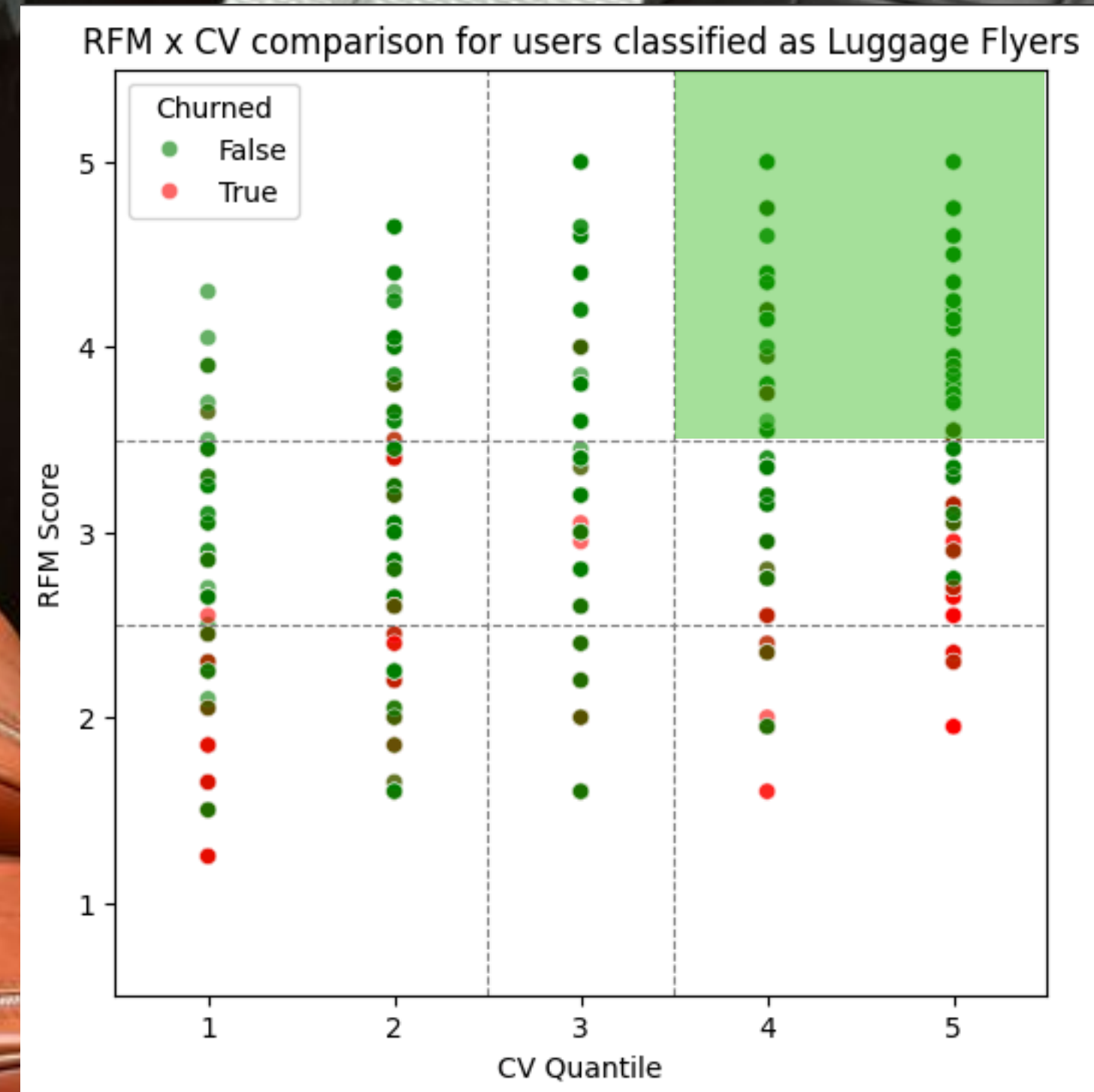
**They don't travel light**

### VIP:

- Goal: retain and maintain
- Free checked bag / Free hand carry capacity increase
- Home-Airport-Hotel transport assistance

### Core:

- Goal: Encourage higher-value trips
- Already booked trips at minimum 50th percentile, so frequency is not the issue.
- Free checked bag for minimum X bags checked. (e.g. buy 2 get 1)
- Free hand carry capacity increase for higher tier trips / for bundle bookings.



# Luggage-heavy Frequent Flyers

899 Users

# Personas for Cultivation

Traditional Segmentation

A large bundle of cinnamon sticks, tied with dried banana leaves, is shown against a grey background. The bundle is rectangular and composed of many individual cinnamon sticks. The ties are made of dried banana leaves, which are light brown and have a fibrous texture. The cinnamon sticks are a warm, reddish-brown color.

Lower than average proportion of  
trips with both flights and hotels  
booked

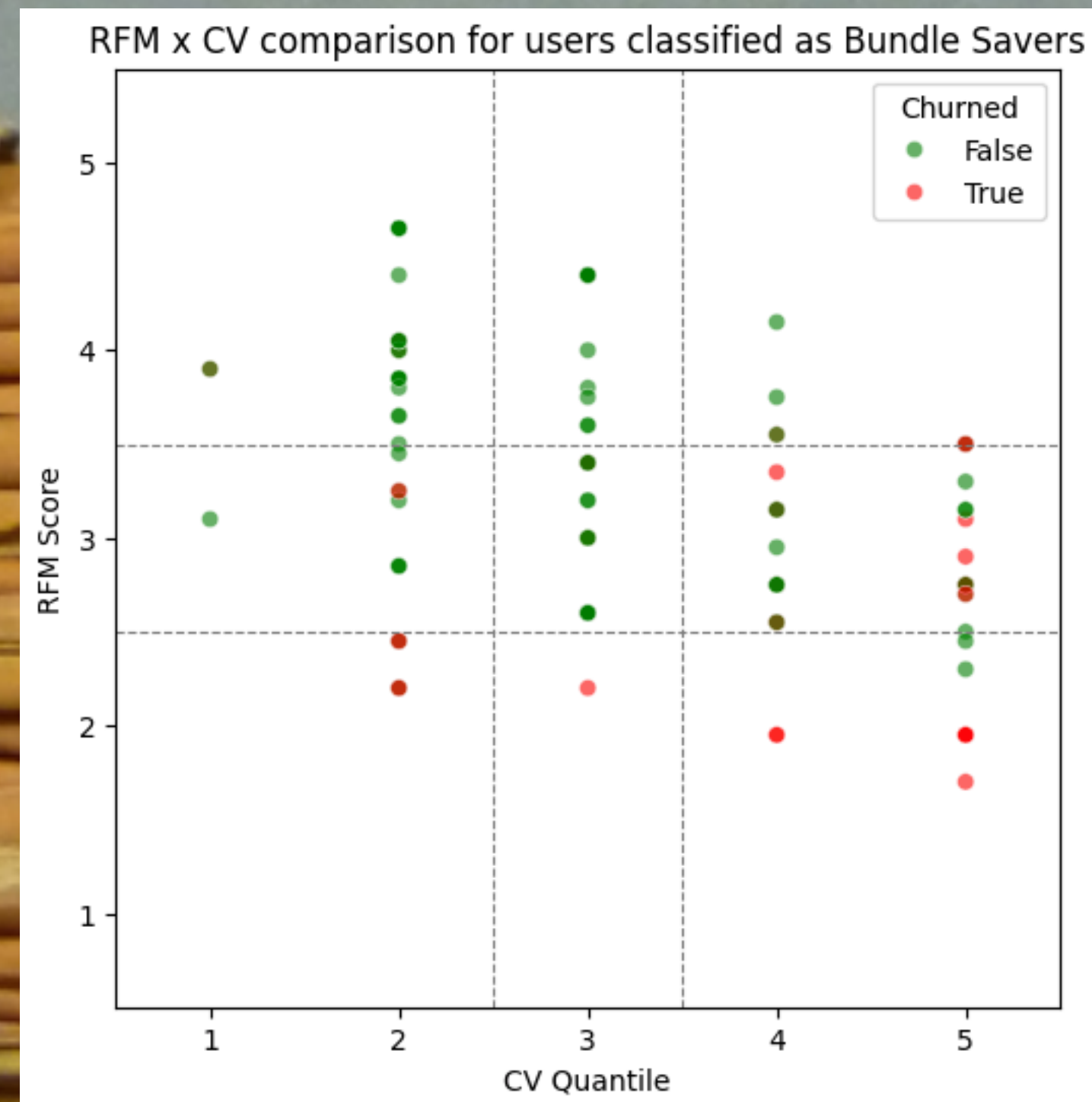
Short average trip length  
Mid-high total spend, but not  
extravagant

# Bundle Savers

Try both flights and hotels

# Bundle Savers

110 Users



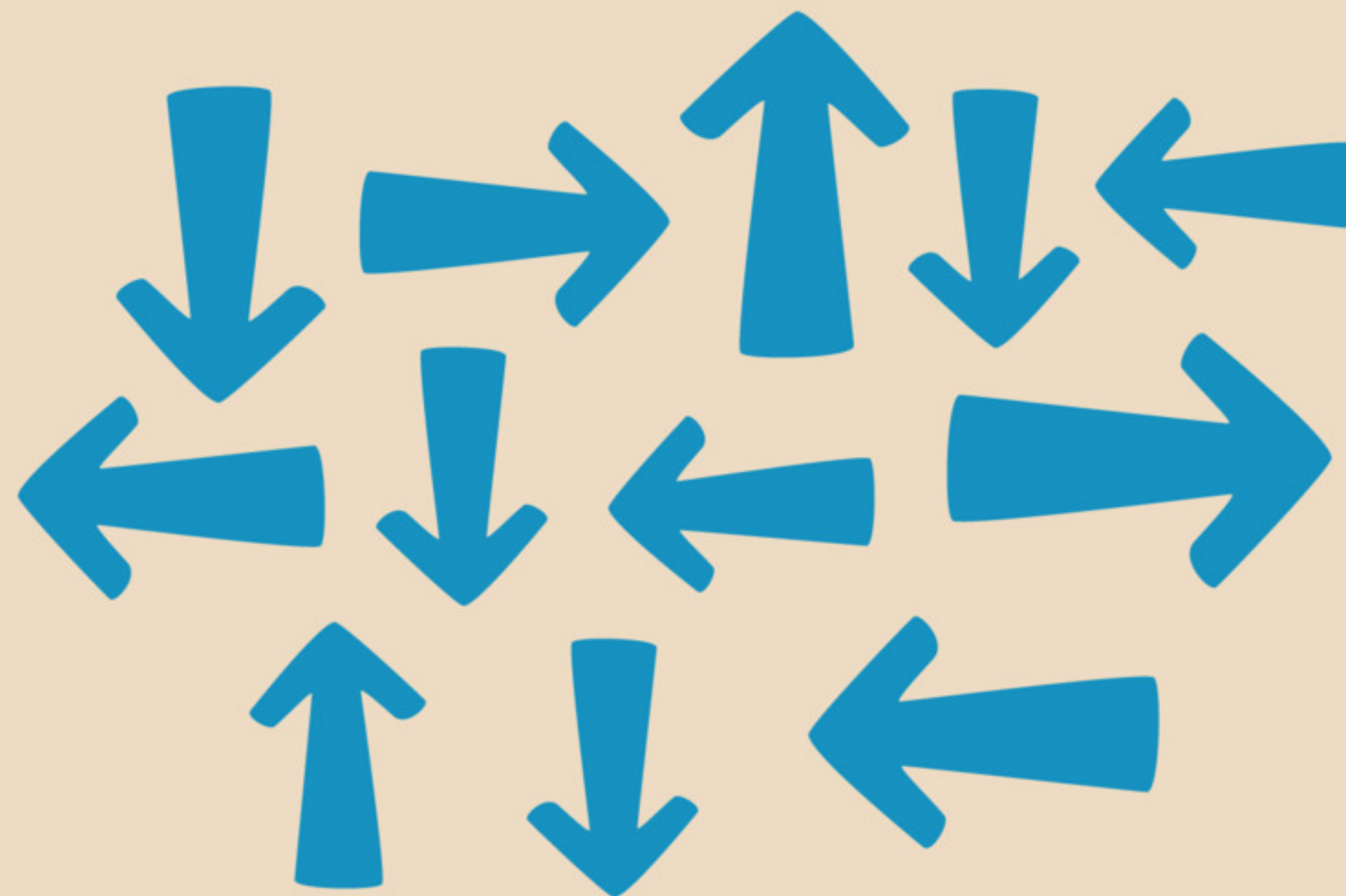
Lower than average proportion of trips with both flights and hotels booked

Short average trip length

Mid-high total spend, but not extravagant

Perk: 1 free night with flight (minimum X nights at mid-high tier hotel, roundtrip)

- 1 extra night is meaningful for shorter trip length
- Decent hotel more attractive than dorm/hostel



Booked only one trip  
Fairly active on the platform  
Spends a lot of time and clicks a lot  
every session



# Uncertain Shoppers

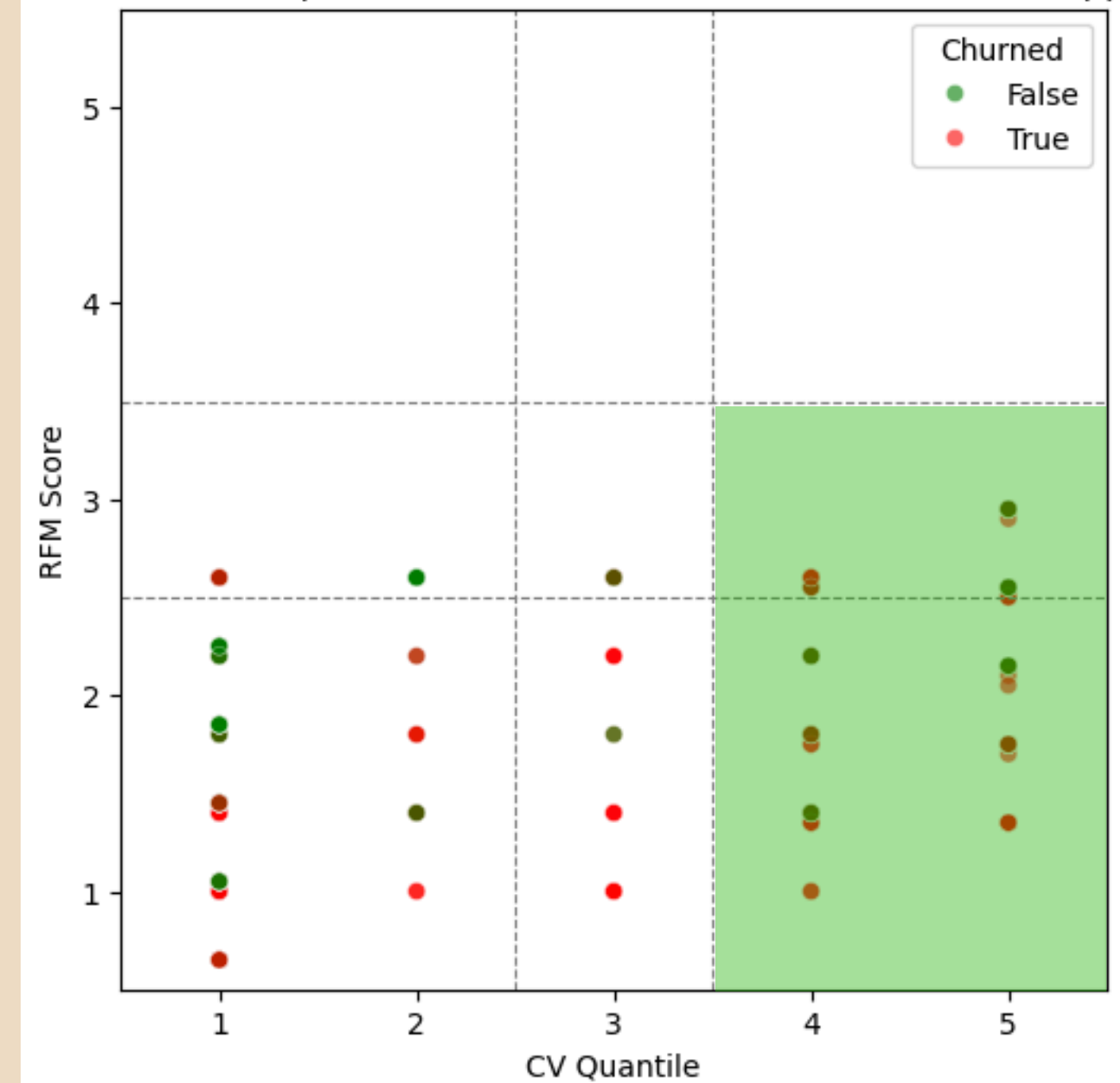
Nudge them to decide

# Uncertain Shoppers

110 Users



RFM x CV comparison for users classified as Uncertain Shoppers



## Growth Potential

- Goal: Reactivate because most have churned
- Free cancellation, price match guarantee, personalized emails with price comparisons and summaries for routes they research

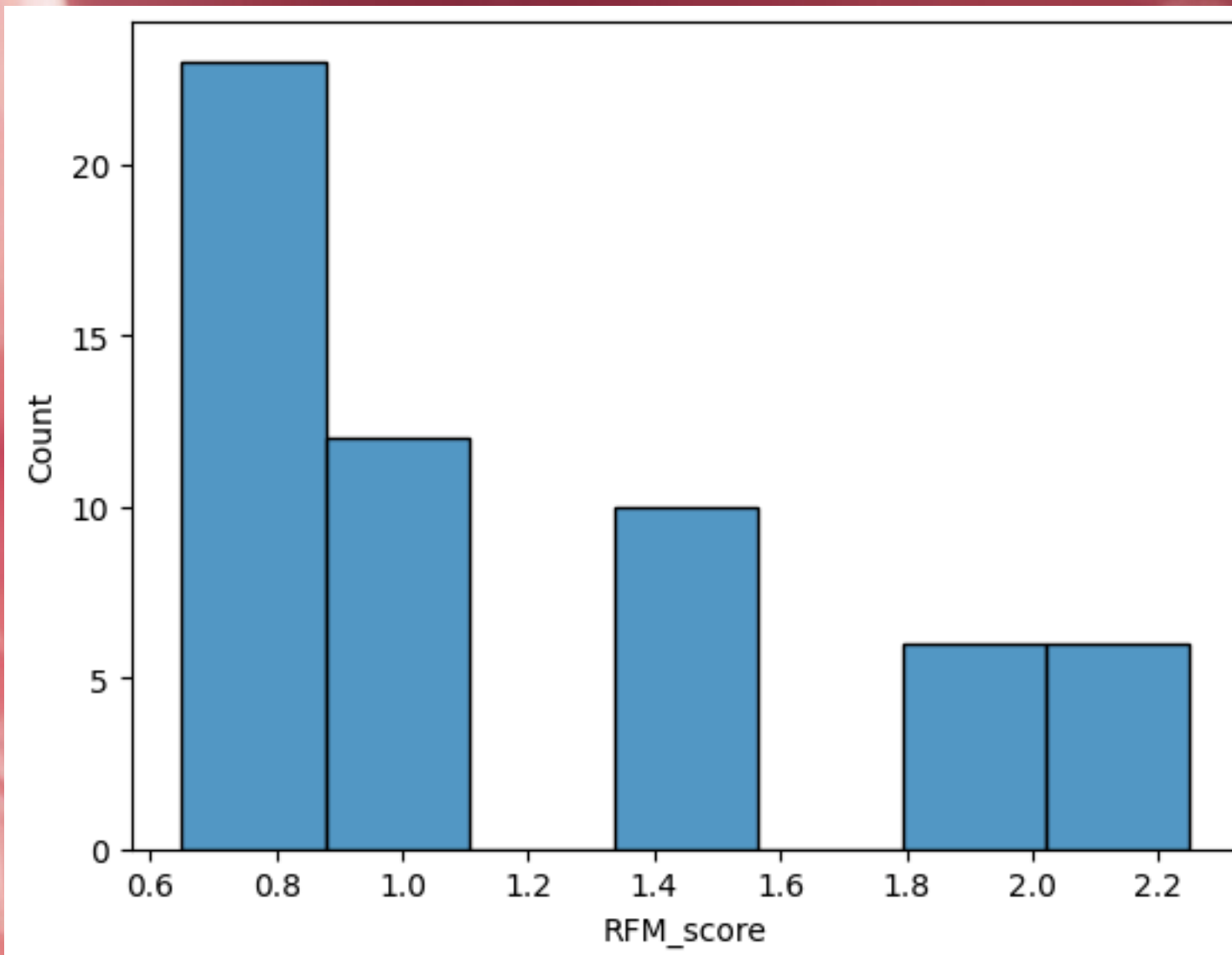


Never booked a trip before  
High average session length and  
clicks per session

# The Uninitiated

The first hit is for “free”

Never booked a trip before  
High average session length and  
clicks per session



Perk: 1\$ flight and 1\$ for one night  
(with minimum number of nights)  
- Overcome the “first purchase” hurdle to  
make future purchases easier on the  
psyche.

# The Uninitiated

57 Users

# Personas found by the Machine

Strong Personas

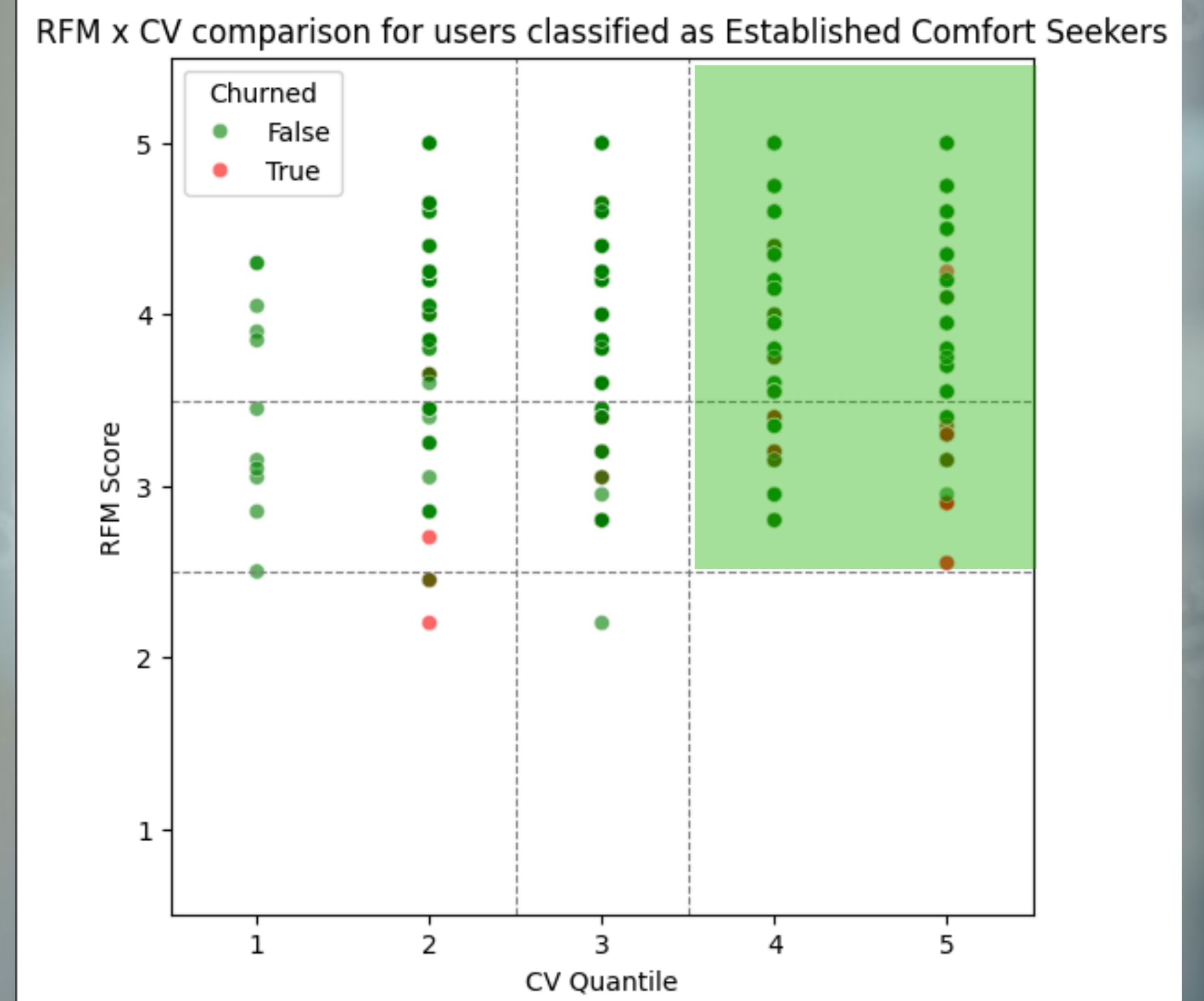
# Established Comfort-Seeker

Enjoying life

Normally distributed around 42.5 years old  
Slightly lower price point than luxury  
travelers, and also does a little more  
research (higher clicks)  
Often books hotel and flights together.

# Established Comfort-Seeker

746 Users



## VIP:

- Goal: retain and maintain
- Offer free/discounted experiences together with the trip bundles that are fitting for middle-aged people (e.g. spa, boat, museum, wine-tasting)

## Core:

- Pretty high RFM across the board
- Goal: try to move to higher Customer Value
- Exclusive discounts on higher-tier destinations with bundled experiences.

The background image is a collage. On the right, a man with short brown hair and a light beard, wearing a light blue button-down shirt and a black backpack, is smiling at the camera while holding a bowl of food and a glass of beer. On the left, a large white bowl is filled with a meal consisting of fried chicken, shredded yellow cheese, green onions, and other ingredients. A yellow text box is overlaid on the top left of the food bowl.

Lower-budget and fewer trips mirror of  
Established Comfort Seekers  
Centering around 41.9 years old  
Also books mostly flights and hotels  
together

Similar strategy with Established Comfort  
seekers, just at a lower price point.

**Budget Comfort-Seeker**

Enjoying life on a budget

### VIP:

- Goal: retain and maintain
- Offer free/discounted experiences together with the trip bundles that are fitting for middle-aged people (e.g. spa, boat, museum, wine-tasting). Free hotel meal

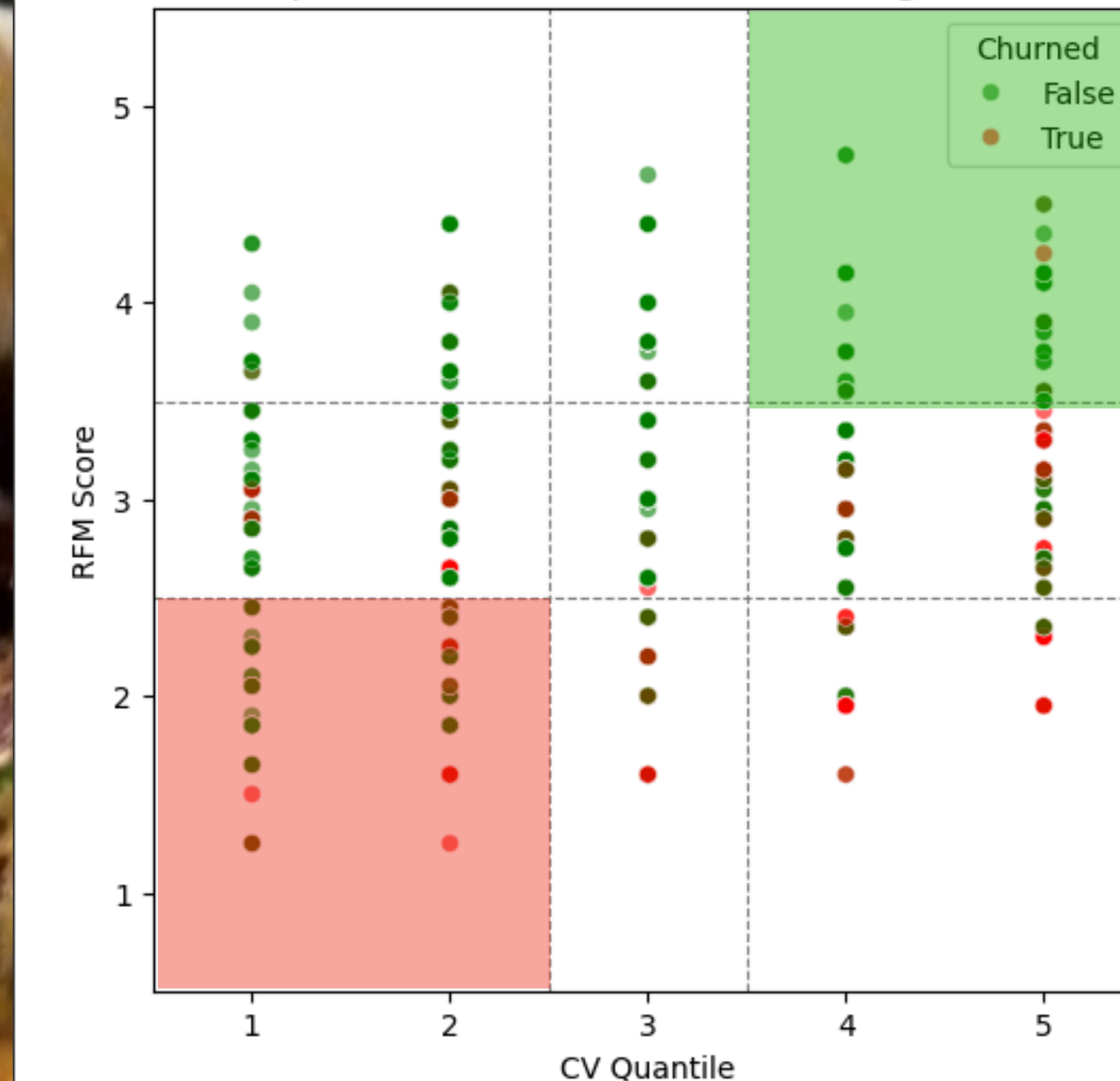
### Core:

- Goal: try to move to higher Customer Value
- Exclusive discounts on higher-tier destinations with bundled experiences. Free hotel meal

### At Risk:

- Goal: Low-cost activation / deprioritize
- Deal alerts / reminder emails

RFM x CV comparison for users classified as Budget Comfort Seekers



# Budget Comfort-Seeker

966 Users

# Personas found by the Machine

For Cultivation



# Bundle Savers Lite

Try both flights and hotels

Lower Recency/Frequency mirror of  
Bundle Savers but with longer  
average trip length  
Often books only either hotel or  
flight  
Some discount sensitivity  
Potential for upsell

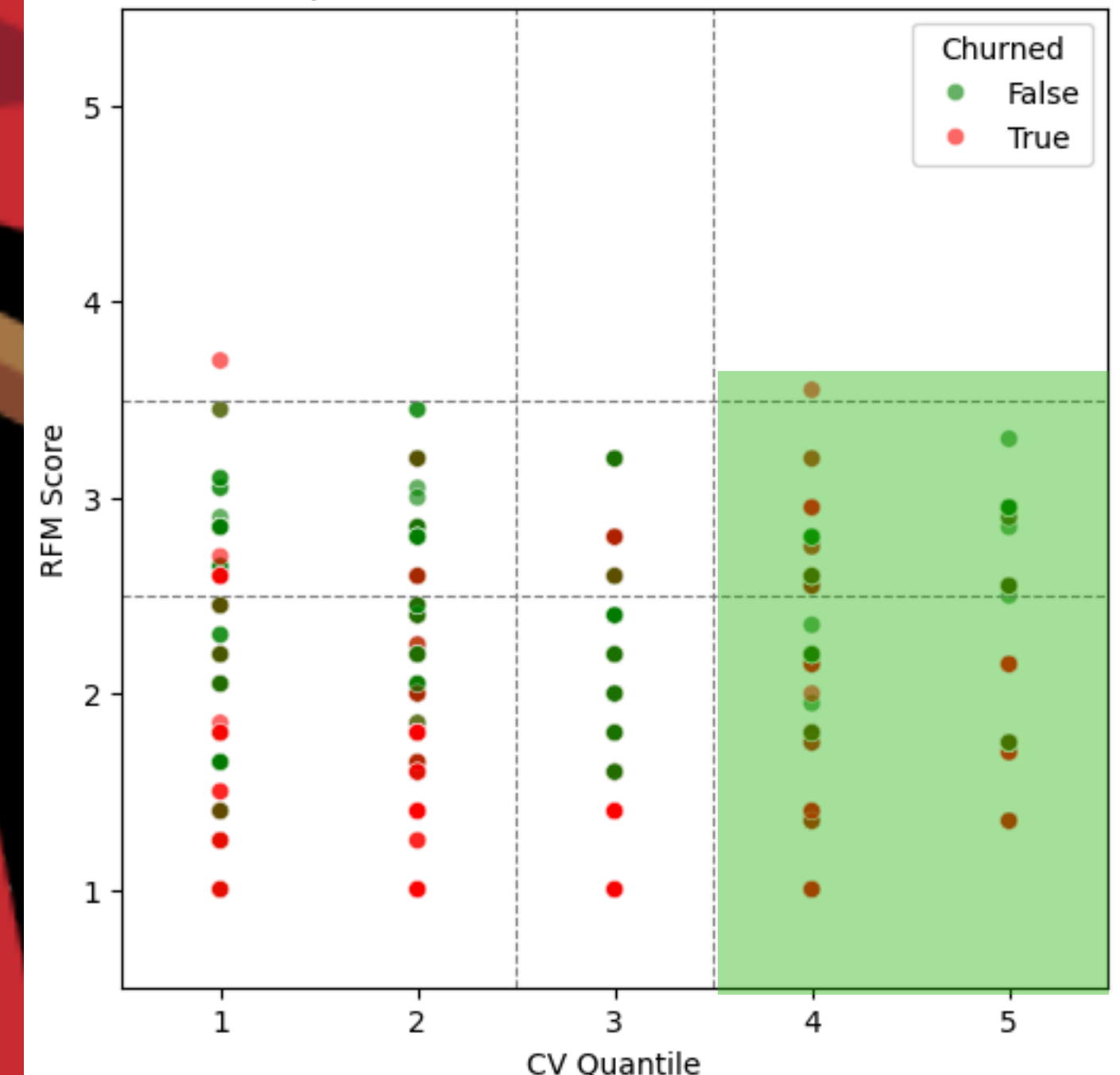
# Bundle Savers Lite

110 Users

## Growth Potential

- Goal: Reactivate the churned, improve frequency.
- Flash deals for hotel and flight bundles with attractive discounts

RFM x CV comparison for users classified as Bundle Savers Lite





Almost all have booked only one trip, and only booked a flight Above average booking lead time / planning horizon  
Has some discount sensitivity  
Have been away from the platform the longest among the clusters

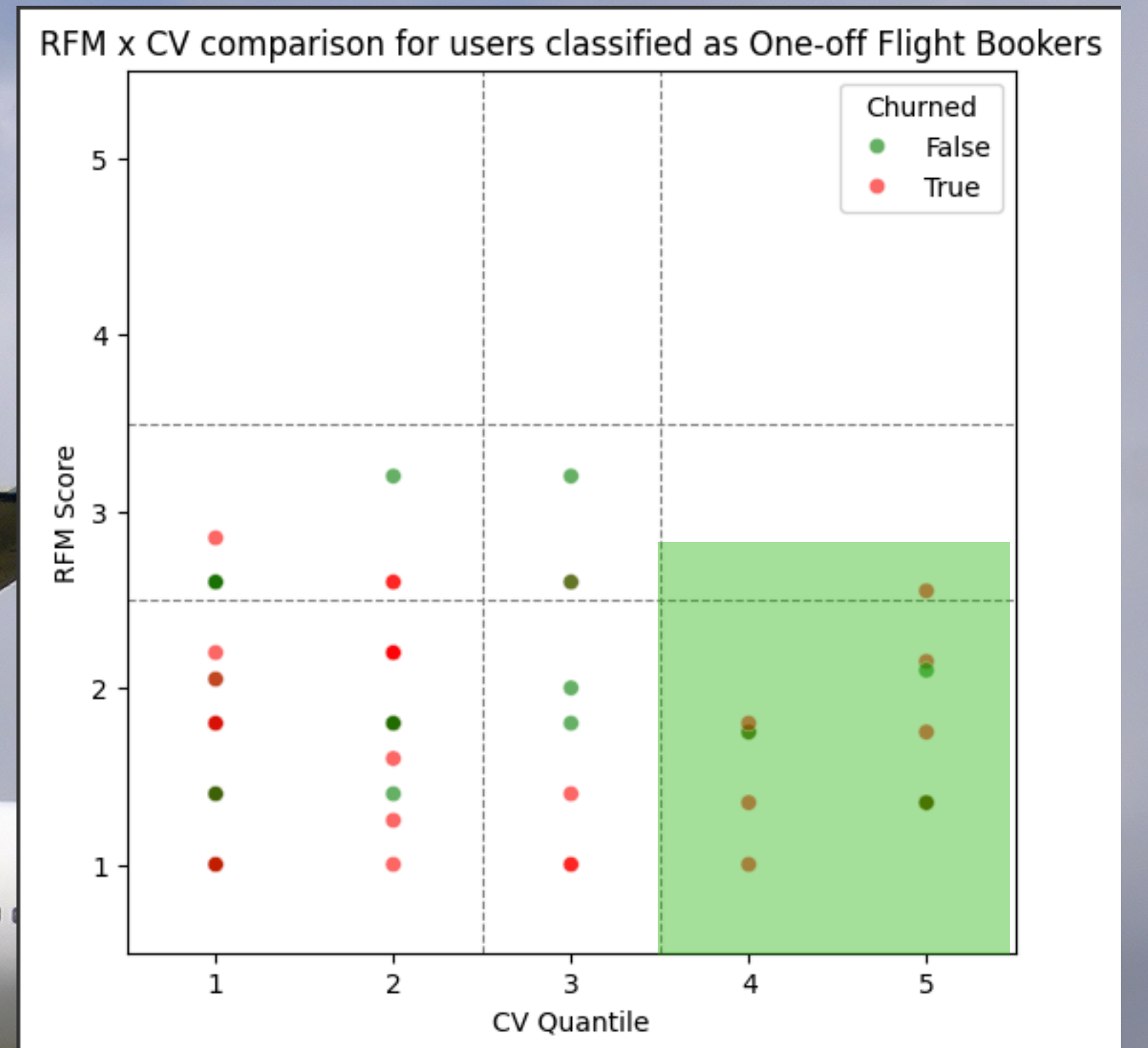
# One-off Flyers

They flew but once



# One-off Flyers

64 Users

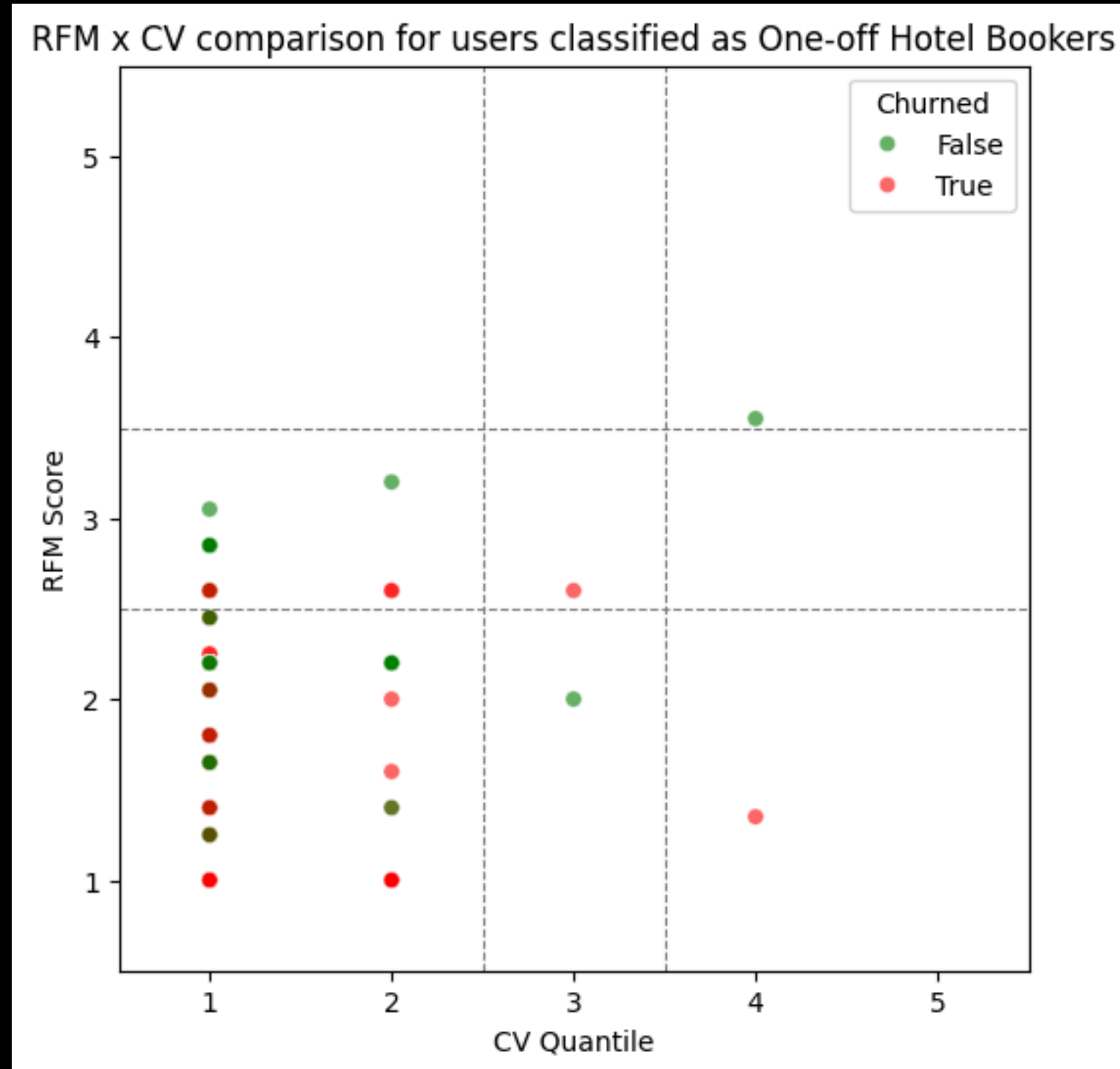


## Growth Potential

- Goal: Reactivate the churned, improve frequency.
- Flash deals / limited time vouchers for airplane seat upgrades (e.g. voucher for leg room upgrade, economy to business class), free cancellation to lean into early booking behavior

# Personas found by the Machine

**Weak / Low Priority**

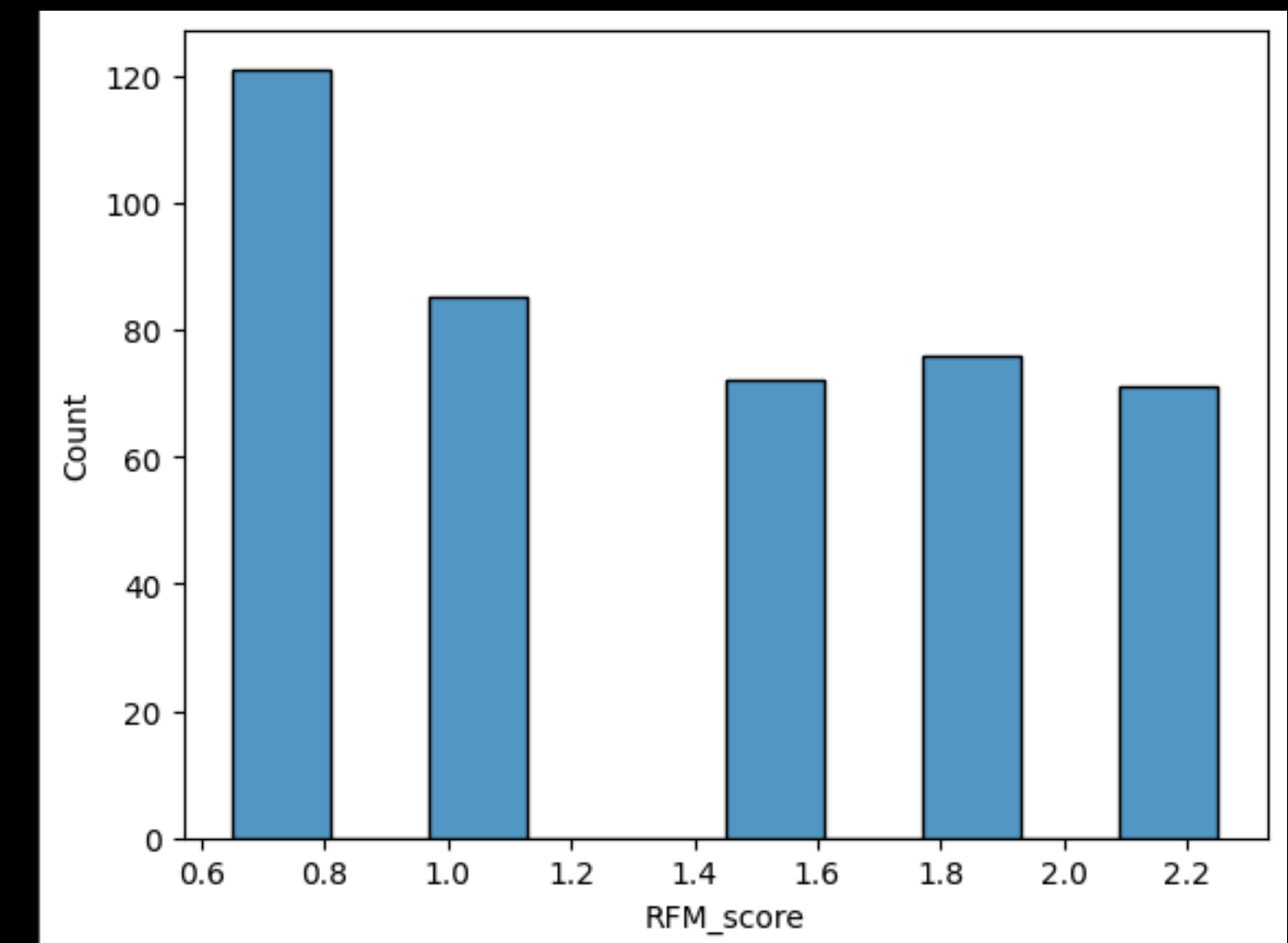


Almost all have booked only one trip, and only booked a hotel  
 50% of the group booked trips of 6 days or less  
 Has some discount sensitivity  
 Overall low RFM and CV

## One-off Hotel Bookers

190 Users

Never booked trips, like the Uninitiated  
 Did not seem engaged during their sessions  
 (lower session duration, lower clicks)  
 Have been inactive for a longer time



## The Uninterested

425 Users

# Next Steps / Potential for Improvement

# Next Steps / Potential for Improvement

- Run targeted marketing campaigns based on segmentation
- Determine most-effective perk for intersections of segments
- Try Unsupervised ML on the same limited parameters

# Bonus

# RFM Analysis

- Days since last session - recency
  - Weighted 0.4. Recency might signal likelihood of booking again soon
- Total spend - monetary
  - weighted 0.35. Big spenders are important but they may likely be the "whales" or VIP users, not the bulk of the user base.
- Total trips - frequency
  - weighted 0.25 because not booking often does not necessarily mean they are low-value

# Customer Value (CV)

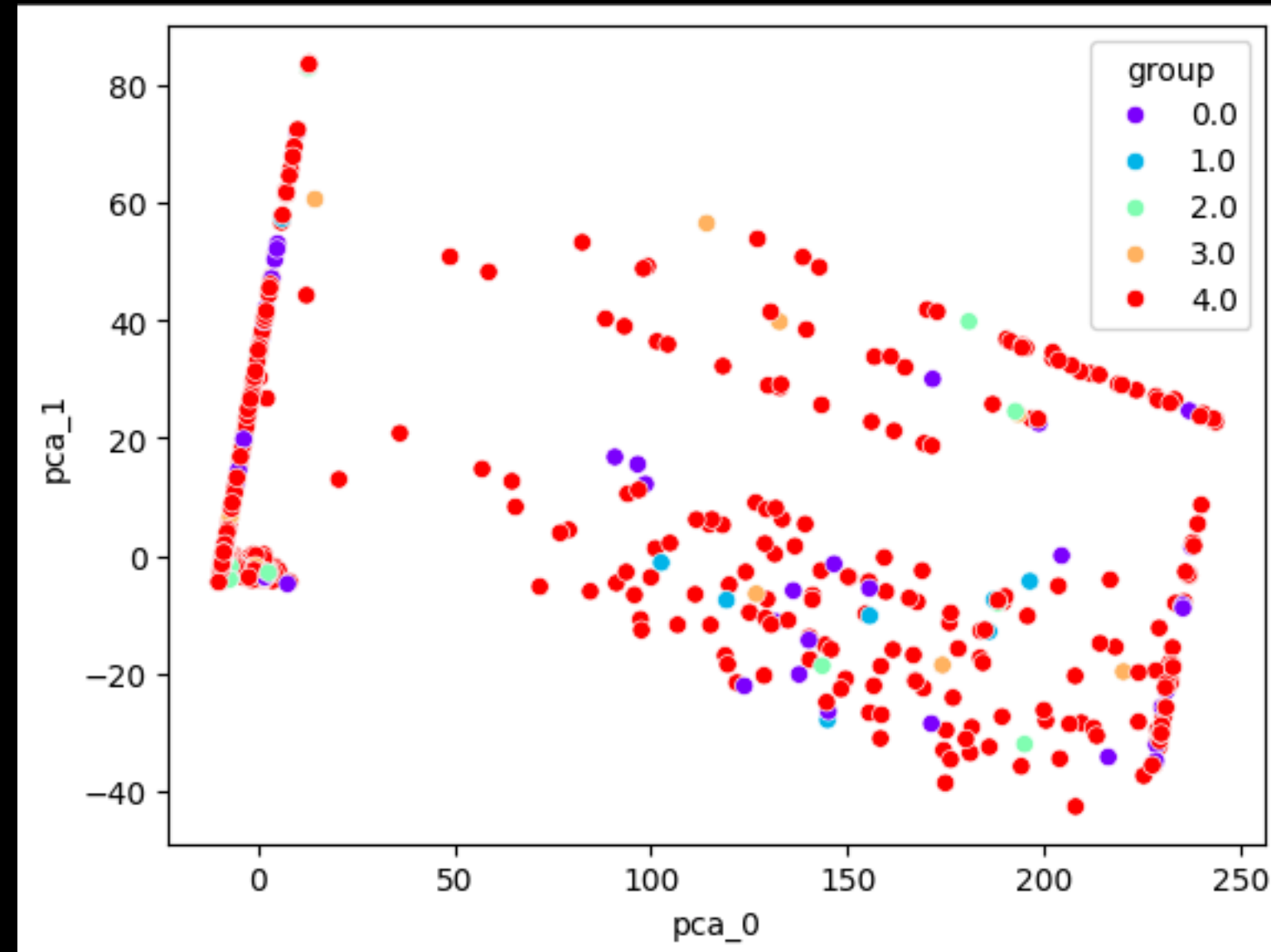
- Calculated per user
- $\text{Average Spend Per Trip} / \text{Average Number of Trips across all users}$

# Churn Flagging

- Users whose last booking is twice or more as long as the average inter-purchase duration
- Average inter-purchase duration: ~45 days

# ML Clustering

## Lots of Correlations



- 95% variance between users captured within three Principal Components

# Cluster Intersections

## Traditional Segmentation

